(1942-2017)

BANK OF THAILAND

MONETARY POLICY REPORT

December 2018

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
December 2018

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Kanit Sangsubhan	Member
Mr. Subhak Siwaraksa	Member
Mr. Somchai Jitsuchon	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2018. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Executive Summary

Monetary Policy Conduct in the Fourth Quarter of 2018

In the Committee's view, the Thai economy continued gaining traction and expanded at the level consistent with its potential. Headline and core inflation were projected to be broadly in line with the previous assessment. Financial stability remained sound overall, but it was deemed necessary to monitor risks that might lead to the build-up of vulnerabilities in the financial system. **The Committee weighed various factors in determining the most appropriate course of monetary policy and voted 4 to 3 at the meeting on November 14, 2018 to maintain the policy rate at 1.50 percent and voted 5 to 2 at the meeting on December 19, 2018 to raise the policy rate by 0.25 percentage point to 1.75 percent.** At the latest meeting, most Committee members viewed the continuity of economic expansion sufficiently strong. The need for currently extra accommodative monetary policy as in the previous period had thus reduced and the policy rate should be raised to curb financial stability risks and to start building policy space. Moreover, the policy rate at 1.75 percent would still be conducive to continuous economic growth. Nonetheless, two members viewed that risks and uncertainties on the external front heightened and could affect Thailand's economic growth in the period ahead and thus saw the need to assess the clarity of such impacts on the sustainability of domestic growth momentum. In addition, the implemented macroprudential measures had addressed certain risks to financial stability to some extent.

Looking ahead, the Thai economy was projected to continue to gain traction despite the slowdown in external demand. The Committee viewed that accommodative monetary policy would remain appropriate in the period ahead. The policy rate increase would be gradual and not in a continuous manner as in the past. The Committee would thus closely monitor developments of economic growth, inflation, and financial stability, together with associated risks, in deliberating appropriate monetary policy in the period ahead.

Assessment of the Economic and Financial Outlook as the Basis for Policy Formulation

1. Global Economy

The global economy was expected to expand at a slower rate due to lower-than-expected outturns in several countries during the third quarter of 2018. However, most Thailand's trading partners would still operate above potential. While there was additional momentum from the U.S. economy due to greater fiscal stimulus expected to take effect in 2019, growth momentum from other countries would slow down because domestic demand was affected by tightening financial conditions, less fiscal stimulus, as well as uncertainties regarding the U.S. trade protectionism measures and political issues in the euro area. **The Committee thus revised down the growth projection of Thailand's trading partners to 3.6 and 3.4 percent in 2018 and 2019, respectively. The projections would be subject to increased downside risks** mainly due to an increased likelihood of the no-deal Brexit. Other risks that warranted monitoring included uncertainties pertaining to the trade protectionism measures, geopolitical risks, and China's financial stability concerns.

Most central banks started to reduce the degree of accommodative monetary policy. The U.S. Federal Reserve (Fed) still gradually normalized its monetary policy. The European Central Bank (ECB) would end new bond purchases at the end of 2018 as announced, but would roll over matured bonds and was expected to maintain its interest rate until the second half of 2019. The Bank of Japan would likely maintain their short-term and long-term target rates for some time. On the other hand, some regional central banks raised their policy rates in the fourth quarter of 2018, for instance, the Bangko Sentral ng Pilipinas, Bank Indonesia, and the Bank of Korea. Looking ahead, other countries in the region were expected to gradually normalize their monetary policy to be consistent with the economic growth outlook and to preserve price and financial stability.

Emerging markets (EMs) experienced capital outflows, as investors viewed the prospect of the Fed's rate hikes to be more certain and there were concerns over intensifying trade protectionism measures by the U.S. and heightened political uncertainties in Europe. Nonetheless, capital flows partly returned to EMs after such concerns somewhat alleviated. Looking ahead, global financial markets would likely remain highly volatile. EM capital flows could fluctuate given uncertainties surrounding the Fed's future rate hikes, possibly intensifying trade protectionism measures after the announcement of temporary suspension of new tariffs, and uncertainties pertaining to Brexit negotiations which warranted close monitoring.

2. Financial Conditions and Financial Stability

Thailand's financial conditions remained accommodative, although overall government bond yields increased at the beginning of the fourth quarter before falling afterwards. This is due to greater demand from foreign investors, the lower-than-expected outturns of Thailand's economic figures in the third quarter, and investors' concerns over uncertainties on the external front. The new loan rate remained at a low level. Private credit expanded in both business and household sectors. Businesses continued to seek funding through both debt and equity instruments. The Thai baht depreciated against the U.S. dollar from the end of the previous quarter due to investors' concerns over uncertainties on the external front. The baht depreciated at a larger extent as compared with other regional currencies. This was due partly to some corrections after previously appreciated faster than other regional currencies. The real effective exchange rate (REER) depreciated.

Financial stability remained sound but there remained pockets of risks that warranted monitoring. These included, first, the search-for-yield behavior, in particular by saving cooperatives and large corporates, which persisted in the prolonged low interest rate environment and could lead to underpricing of risks. Second, the property sector was subject to accumulated vulnerabilities in the mortgage loan market and oversupply in the real estate market. The markets would likely adjust to the implemented revision in the macroprudential measure on mortgage loans during the period prior to the effective date. Housing demand would likely rise as a consequence of accelerated purchasing of mortgage loan borrowers for second or more homes and the developers would adopt more proactive marketing strategy. Third, despite household debt slowly declining, it remained at a relatively high level and the debt serviceability of some groups of households and small corporates continued to deteriorate.

3. Economic and Inflation Outlook

The Thai economy was projected to record slower growth at 4.2 and 4.0 percent in 2018 and 2019, respectively, lower than the projection in the previous *Monetary Policy Report*. Growth drivers stemming from external demand were expected to moderate as merchandise and services export growth would be lower than expected. Meanwhile, domestic demand would remain a key growth driver, especially on the back of sustained private consumption growth.

Merchandise exports were expected to slow down in tandem with global demand. This was partly due to export prices which would decline following a fall in crude oil prices and a temporary factor pertaining to natural disasters in Thailand's trading partners. The value of merchandise exports was projected to grow at lower rates of 7.0 and 3.8 percent in 2018 and 2019, respectively. The downward revisions were owing to impacts of the U.S. trade protectionism measures on global trade volume and Thailand's trading partner economies. Nonetheless, the redirected orders and the relocation of production base from China to Thailand would be supporting factors in the period ahead. Under the Committee's assessment, although concerns over trade protectionism measures alleviated in the short term, there remained uncertainties which could put pressures on the international trade and investment. This would both directly and indirectly affect Thai exports. Thus, it was deemed necessary to closely monitor developments of trade negotiations.

Exports of services were projected to exhibit slower growth due to a gradual recovery in Chinese tourist figures. The projected number of foreign tourists was revised down to 38 and 40 million for 2018 and 2019, respectively. This was owing to a large contraction in Chinese tourist arrivals following more severe and longer-than-expected impacts of the Phuket tour boat sinking incident, coupled with additional effects of Chinese economic slowdown and the renminbi depreciation. However, signs of improvement in Chinese tourists were observed after the government promptly enhanced safety standards in the tourism sector and put in place marketing incentives such as the free visa-on-arrival.

Private consumption was expected to continue expansion in almost all categories, particularly for durable and semi-durable goods. Growth would be partly supported by the waning effect of the first-car scheme, more broad-based growth of non-farm income, and continued improvements in employment in major sectors, as well as support from government policies. However, private consumption growth was expected to moderate somewhat in the period ahead due to the high-base effect of durable goods consumption in 2018. Moreover, elevated household debt would cause households to allocate part of their income for debt repayment. Structural changes in the labor market would limit wage increases. These factors would restrain purchasing power in the period ahead.

The government would help drive the economy despite some slowdown in public spending compared with the previous assessment. Delays in public investment were expected in some projects. Central government investments were revised down due to the delayed approval of land expropriation expenses for the Bang Yai-Kanchanaburi motorway project. Moreover, delays in drafting the master plans under the National Strategy prompted government agencies to defer obligation of funds for some projects. For state-owned enterprises, investments were revised down due to the

deferment of the second phase of the dual-track rail project which was during the process of project priority review. Government consumption growth was projected to be in line with the previous assessment.

Private investment growth projections were largely unchanged, but would gradually improve mainly on the back of machinery and equipment investment. Drivers would come from upgrades to production efficiency and capacity expansion as well as the relocation of production base to Thailand. Such factors would compensate for the moderating investment trend owing to investor concerns over trade protectionism measures as well as delays in public investment. Additional supporting factors for the investment outlook were (1) special measures by the Board of Investment (BOI) to boost investment and (2) the public-private partnership (PPP) in infrastructure investment projects in the Eastern Economic Corridor (EEC). Progress of such investment projects would be monitored going forward.

The inflation outlook was expected to be in line with the previous assessment. Headline inflation was projected to be slightly lower than the previous assessment due to lower crude oil prices despite some offset by higher fuel adjustment surcharge (Ft). Core inflation was expected to slightly increase due to higher bus fares and gradually improving demand-pull inflationary pressures. **The Committee thus projected headline inflation to be 1.1 and 1.0 percent for 2018 and 2019, respectively, and core inflation to be 0.7 and 0.9 percent for 2018 and 2019, respectively.**

Risks to the growth projection were expected to tilt more downward, mainly due to external factors as the probability of no-deal Brexit heightened. Moreover, the possibility that the Thai economy would underperform the baseline forecast could be due to (1) impacts of the U.S. trade protectionism measures, (2) lower-than-expected growth of trading partners, (3) slower-than-expected growth of private consumption due to an uneven distribution of domestic purchasing power recovery, especially for farm income, and (4) lower-than-expected public expenditure due to constraints in budget disbursement and project execution. On the other hand, the possibility that the Thai economy would outperform the baseline forecast could be due to (1) less-than-expected Chinese economic slowdown, and (2) possibly higher-than-expected domestic spending due to infrastructure investment and PPP projects as well as government measures to help stimulate private consumption. Meanwhile, **risks to the forecasts of headline and core inflation were expected to tilt more downward** in line with increased downside risks to growth projections and to oil prices.

1. Global Economy

Major advanced economies were expected to grow with greater divergence on their growth rates. While the U.S. and Japanese economies would likely expand, growth momentum in the euro area was expected to slow down due to domestic factors and global trade slowdown.

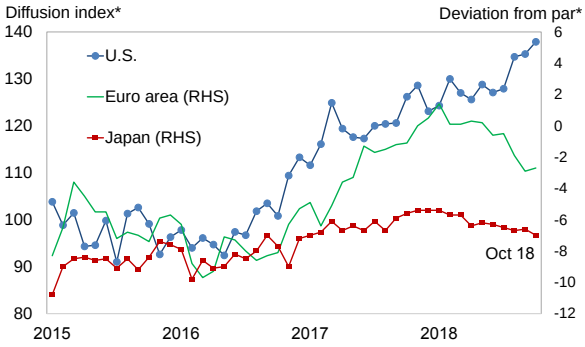
The U.S. economy was expected to continue gaining traction. In the third quarter of 2018, the U.S. economy saw better-than-expected growth as businesses built up inventories from the previously low level. Looking ahead, the U.S. would continue to grow on the back of domestic demand with support from strong labor market, high consumer confidence and policies regarding personal and business income tax cuts. Moreover, increased government spending after the U.S. Congress passed the budget bill for 2019 would be an additional support for the economy in the period ahead. On the other hand, **the euro area was expected to expand moderately**. In the third quarter of 2018, the economy slowed down temporarily due to the new emission testing regime which had some temporary effects on automobile production. In the period ahead, growth momentum was expected to slow down due to further restraint in private sector confidence following political uncertainty in various countries. In addition, global trade volume, which would be moderate, could affect exports in some countries. **The Japanese economy was expected to continue expanding** despite the negative growth rate in the third quarter of 2018 due to temporary factors, namely the natural disasters from storm and earthquake. Nonetheless, domestic demand was expected to recover in the fourth quarter as supported by strong labor market and continued accommodative monetary policy. In the period ahead, private consumption would likely accelerate prior to the consumption tax increase in October 2019, although merchandise exports could be affected by the global trade slowdown (Chart 1.1).

The Chinese economy was expected to slow down due to a series of financial stability measures released previously. Meanwhile, growth momentum of other Asian economies somewhat declined on account of tightening financial conditions, confidence which started to be affected by the U.S. trade protectionism measures, and lower fiscal stimulus in some countries.

The Chinese economy was expected to exhibit slower growth as financial conditions were tightened because banks previously tightened credit standards according to a series of the Chinese government's financial stability measures to curb debt levels in the economy. In addition, the impacts of the U.S. trade protectionism measures would be increasingly prominent in the period ahead despite the recently limited effects on Chinese exports which accelerated before the full implementation of the U.S. tariffs imposed. Nonetheless, the government's growth-supportive measures, including accelerated investment in infrastructure, tax deductions for businesses and households, and injection of more liquidity into the financial system by reducing the banks' reserve requirement ratio (RRR), would help alleviate the adverse impact of the financial stability and trade protectionism measures to some extent.

Asian economies, excluding China and Japan, were expected to slow down. In the third quarter of 2018, several economies recorded slower growth than previously assessed. Domestic demand began to experience some impacts from tightening financial conditions in some countries, slumping confidence following the U.S. trade protectionism measures, and lower fiscal stimulus in some countries. In particular, the Malaysian government postponed some public investment to stabilize fiscal deficit and the Indonesian government also delayed in some state investment to reduce the size of current account deficit. Nonetheless, domestic demand in Asian economies were expected to expand in the period ahead supported by robust economic fundamentals, especially strong labor markets and upward wage adjustments in many countries. Meanwhile, merchandise exports were expected to register slower growth in line with the global economic outlook, coupled with indirect effects of trade protectionism measures on global supply chain.

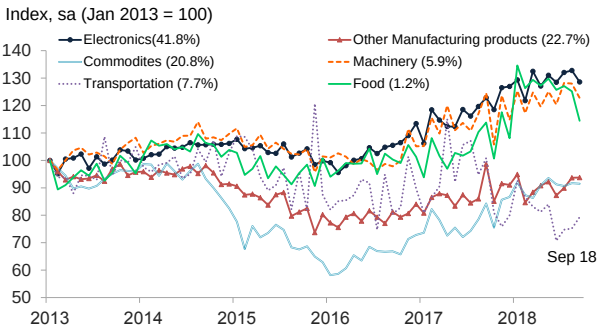
Chart 1.1 Consumer confidence remained high in the U.S., but continued to decline in the euro area and Japan



Note:*Euro area's par = 0 and Japan's par = 50
Source: Bloomberg

Chart 1.2 Asian exports expanded due to the front-load order before the full implementation of the announced U.S. tariff measures

Asia n exports value* classified by product categories



Note: *Asian exports include Hong Kong, Taiwan, Korea, Malaysia and Singapore.
( ) denotes share of total exports in 2017
Commodity-related products include crude oil, metals, chemicals, rubber, and vegetable oil.
Other manufacturing products include textile, papers, furniture, footwear and miscellaneous
Source: CEIC

The growth projection for Thailand's trading partners was revised downward, with greater downside risks to the baseline projection.

Thailand's trading partner economies were expected to record slower growths than previously projected in the previous *Monetary Policy Report*. The downward revision was due to lower-than-expected growth outturns during the third quarter of 2018 in many countries. Going forward, most trading partner economies would continue expanding above their potential levels, coupled with an additional momentum from the U.S. economy due to additional fiscal stimulus in 2019. Meanwhile, growth momentum of other countries would slow down because domestic demand was affected by tightening financial conditions, lower fiscal stimulus, as well as uncertainties regarding the U.S. trade protectionism measures and political situations in the euro area. The Committee, thus, revised down the growth projection for Thailand's trading partners from 3.8 and 3.5 percent to 3.6 and 3.4 percent in 2018 and 2019, respectively (Table 1.1).

Under the Committee's assessment, there were increased downside risks that Thailand's trading partner economies would underperform the baseline projection due mainly to an increased likelihood of the no-deal Brexit. Other risks that warranted monitoring included, first, uncertainties pertaining to the U.S. trade protectionism measures despite the recent United States-Mexico-Canada Agreement (USMCA) and the temporary truce on the

U.S.-China trade tensions after the G20 summit. The U.S. and China agreed to suspend new trade tariffs for 90 days in an attempt to mutually reach an agreement. Second, uncertainties regarding geopolitical risks, particularly the denuclearization negotiations between the U.S. and North Korea which has yet to materialize, as well as recurring tensions in the Middle East, could heighten volatility in the global financial and commodity markets, especially through oil prices, and would, in turn, affect the real economy. Moreover, there remained risks pertaining to political uncertainties in Europe, particularly the Brexit negotiations. Although the draft of UK's withdrawal agreement was recently approved at the EU summit, the U.K. parliament was expected to vote on the deal later. Third, financial stability concerns in China persisted. Despite the government's intensifying financial stability measures, there remained challenges in balancing between promoting economic growth and preserving financial stability.

Table 1.1 Assumption on trading partner growth

Annual change (%YoY)	Weight (%)	2017*	2018	2019
United States	14.9	2.2	2.9 (2.8)	2.6 (2.3)
Euro area	10.0	2.5	1.9 (2.1)	1.6 (1.8)
Japan	13.6	1.9	0.8 (1.2)	1.0 (1.0)
China	15.7	6.9	6.6 (6.6)	6.2 (6.2)
Asia (excluding Japan and China)**	37.4	4.6	4.2 (4.3)	3.9 (4.1)
Total***	100	3.9	3.6 (3.8)	3.4 (3.5)

Note: *Outturn
**Weighted by a share of Thailand's total exports to 7 trading partners in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8.0%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%), and the Philippines (3.7%)
***Weighted by a share of exports from Thailand to 13 trading partners in 2014 (including the United Kingdom and Australia)
( ) as reported in Monetary Policy Report, September 2018

Most central banks started to normalize monetary policy.

In 2018, the U.S. Federal Reserve (Fed) raised the federal fund rate four times in total, in line with expectations, on the back of robust economic growth and continued tightening labor market. In 2019, the Fed was expected to raise the policy rate twice instead of three times as projected in the previous assessment. The revision was due to heightened global economic uncertainty, financial market volatility, and the policy rate being closer to the neutral rate. Going forward, the Fed's monetary policy would be particularly flexible and data-dependent. Meanwhile, the European Central Bank (ECB) would end new bond purchases at the end of 2018 as announced, but would roll over matured bonds and was expected to maintain its policy rate until the second half of 2019. On the other hand, some regional central banks raised their policy rates in the fourth quarter of 2018, namely the Bangko Sentral ng Pilipinas, Bank Indonesia and the Bank of Korea (Box: *The recent monetary policy normalization: international perspectives*).

The recent monetary policy normalization: international perspectives

Central banks around the world have implemented ultra-accommodative monetary policy since the outburst of the global financial crisis in 2008. However, as the global economy continued to show a strong recovery, many central banks began to normalize their monetary policy by reducing the degree of policy accommodation. Most importantly, the Federal Reserve (Fed) started raising its policy rate for the first time in December 2015 and has gradually hiked the rate since then. In 2018, the policy rate was raised four times to 2.25 - 2.50 percent. The Fed's policy rate hikes then resulted in an upward trend for global interest rates and the tightening of global financial conditions.

Central banks in Asia also began reducing the degree of monetary policy accommodation during 2017 - 2018 owing to several reasons. The Bank of Korea, for example, started to increase its policy rate for the first time in November 2017 due mainly to financial stability concerns. Against the backdrop of robust economic growth, the Bank Negara Malaysia (BNM) raised its policy rate for the first time in January 2018 to mitigate risks resulted from the prolonged low interest rate. The Bangko Sentral ng Pilipinas (BSP), meanwhile, had its first policy rate hike in May 2018 to preserve price stability after inflation had risen above the 2-4 percent inflation target. Bank Indonesia also raised the policy rate for the first time in May 2018 to alleviate pressures from capital outflows and currency depreciation. The Bank of Thailand began to raise its policy rate in December 2018, later than other countries. This was partly thanks to strong external stability, as reflected by the high level of international reserves and the continued current account surplus, which allowed the monetary policy conduct to be more independent of external factors and greater considerations on domestic factors. Going forward, other regional countries are expected to gradually normalize their monetary policy in line with the economic outlook, as well as to preserve both price stability and financial stability.

Table 1: The policy rates increased in various countries after the 2008 Global Financial Crisis

Country	Date of the first policy rate hike	The initial policy rate (%)	Accumulated policy rate hike (%)	The current level of policy rate (%)
The United States	16 Dec 2015	0 – 0.25	2.25	2.25 – 2.50
The European Union	Unchanged	0	-	0
Japan	Unchanged	- 0.1	-	- 0.1
South Korea	30 Nov 2017	1.25	0.50	1.75
Malaysia	25 Jan 2018	3.00	0.25	3.25
Philippines	10 May 2018	3.00	1.75	4.75
Indonesia	17 May 2018	4.25	1.75	6.00
Thailand	19 Dec 2018	1.50	0.25	1.75

Notes: Data as of 19 December 2018.

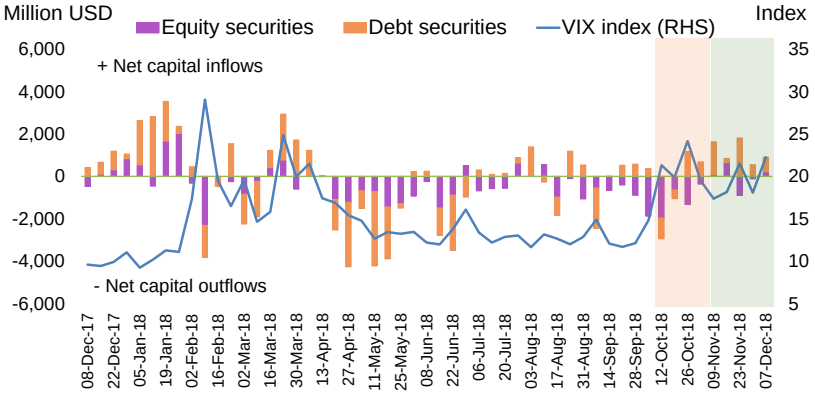
Capital flow experienced volatility due to anticipation of the Fed's rate hike, concerns over the U.S. trade protectionism measures, and political uncertainty in Europe.

In October 2018, emerging markets (EMs) experienced capital outflows, particularly in equity markets, as investors viewed the prospects of the Fed's rate hike to be more certain after the Fed's chairman and some Federal Open Market Committee members (FOMC) expressed views in support of a rate hike. Moreover, investors were concerned about the U.S. trade protectionism measures and heightened political uncertainty in Europe, especially regarding the Brexit negotiations and Italy's budgetary process. Such concerns were reflected by rising VIX[1] index (Chart 1.3). However, capital flows partly returned to EMs in November 2018 after such concerns alleviated as trade negotiations between the U.S. and China exhibited some positive signs and the Brexit negotiation showed some progress, resulting in somewhat declining VIX index. The EM bond markets continued to experience capital inflows, partly due to country-specific factors. For instance, Bank Indonesia continuously raised its policy rate while Thailand's economic fundamentals remained strong and a policy rate hike was anticipated.

Looking ahead, global financial markets would remain highly volatile. Capital flows in EMs could experience volatility as uncertainties loomed on the future U.S. rate hikes, possibly intensifying trade protectionism measures after the announcement of temporary truce of new trade tariffs, and the Brexit negotiations which warranted close monitoring.

Chart 1.3 In October 2018, EMs experienced capital outflows due to market expectations on Fed rate hike, concerns over trade protectionism measures, and political uncertainties in Europe. Capital flows partly returned to EMs in November after concerns subsided.

Net capital inflows to EMs* (weekly) and VIX index



Note: *EMs include Thailand, Indonesia, India, South Africa, and Turkey
Sources: Bloomberg and Institutional Institute of Finance

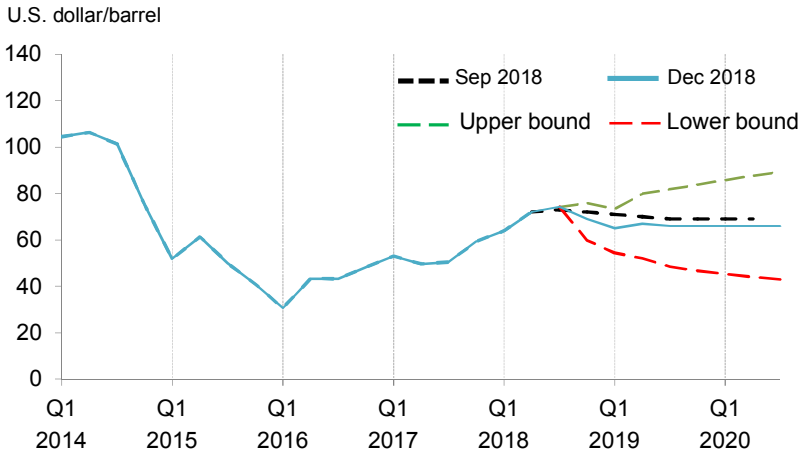
[1/] VIX Index is a measure of the stock market volatility implied by S&P 500 index options

Crude oil prices in the fourth quarter of 2018 declined from the previous quarter. This is due to alleviated concerns over oil supply after the U.S. temporarily lifted some sanctions on Iran and investors' concerns over the global economic outlook that might exhibit a further slowdown.

In the fourth quarter of 2018, the Dubai crude oil prices declined from the previous quarter. Concerns over oil supply subsided after the U.S. had lifted some sanctions on Iran's oil exports for eight major Iran's oil importing countries, such as China, Japan, South Korea and India, for six months. Besides, the U.S. Department of Energy anticipated the U.S. crude oil production to reach a new-high record level at 11.7 million barrel per day at the end of 2018, owing to the gradual increase in production by shale oil producers. Additionally, concerns that global economic growth could further decelerate would weigh on oil prices. Nevertheless, at the beginning of December 2018, the OPEC and allies reached an agreement to cut crude oil production by 1.2 million barrel per day for six months starting from January 2019. This would help support oil prices to some extent.

The Committee therefore revised down the projection for Dubai crude oil prices throughout the forecast horizon given a slowdown in global economic growth, which would affect demand for crude oil. Meanwhile, the agreement to curtail oil production was expected to raise oil prices to a certain extent, thereby preventing a sharp decline of oil prices in 2019. The Committee thus revised down the projection for oil prices from 70.3 to 69.8 dollars per barrel in 2018, and from 69.8 to 66.0 dollars per barrel in 2019 (Chart 1.4). Nevertheless, **there were greater risks that oil prices would be below the baseline projection**. Such risks included the U.S. trade protectionism measures that might impact global trade volume and global economic growth more than previously assessed, a possibly faster-than-expected increase in the U.S. shale oil production, as well as the cut in oil production by OPEC members and allies which could be lower than the agreement. Meanwhile, upside risks that the oil prices would be above the baseline projection were from geopolitical risks, such as those in Libya and Iraq, which might consequently lower supply of oil.

Chart 1.4 Dubai crude oil prices in 2018Q4 declined from the previous quarter amid easing concerns over supply shortages after the US temporarily lifted some sanctions on Iran, coupled with concerns over global economic growth that could slow down further



2. The Thai Economy

2.1 Recent Developments

The Thai economy continued to gain traction in the third quarter of 2018, mainly driven by domestic demand while external demand slowed down.

In the third quarter of 2018, the Thai economy expanded 3.3 percent from the same period last year. The key driver was accelerating private consumption growth across almost all product categories, particularly durable goods following passenger car sales, on the back of improved purchasing power and robust consumer confidence. Private investment accelerated, driven by construction investment, particularly residential construction. Machinery and equipment investment also continued to expand. Meanwhile, public spending expanded on account of procurement of goods and services and social welfare transfer in-kind. However, spending on public investment slowed down in construction projects while investment by state-owned enterprises continued to expand. With regard to external demand, merchandise exports slightly contracted mainly in agricultural products and gold. Exports of some products, such as washing machines, solar cell panels and integrated circuit boards, declined as a result of direct and indirect impact of the U.S. trade protectionism measures. Nevertheless, the value of merchandise exports in September temporarily declined due to the impact of natural disasters in Japan, Hong Kong and the Philippines. Growth in exports of services slowed down in line with declining number of Chinese tourists following the Phuket tour boat sinking incident in July.

The Thai economy continued to gain further traction **in the fourth quarter of 2018**, as reflected by recent economic indicators. Merchandise exports growth recovered as the impact of temporary negative factors were short-lived. However, exports of some product categories were still affected by the trade protectionism measures. Exports of services showed signs of slowing down mainly due to the decline in Chinese tourists as the Chinese government imposed stricter safety standards on destination countries and China's economic growth showed signs of moderating. Meanwhile, domestic demand continued to support the Thai economy. Private consumption indicators recording robust growth thanks to higher household income and employment. Private investment indicators also showed a continued expansion. Public spending expanded on the back of both procurement of goods and services and compensation of civil servants, while public investment expanded on account of investment by state-owned enterprises such as the State Railways of Thailand, the Metropolitan Electricity Authority, and the Provincial Electricity Authority.

The continued economic expansion extended the positive spillovers to the labor market, supporting improvements in purchasing power of households.

The continued economic expansion benefited employment and household income which supported the improvements in purchasing power of households (Chart 2.1 and 2.2). Employment of **non-farm households** recorded healthy growth in both the construction and manufacturing sectors while employment in the services sector declined but still remained at a high level. Going forward, the U.S. trade protectionism measures could impact employment in some industries, such as automobiles and electronics. However, the companies in these

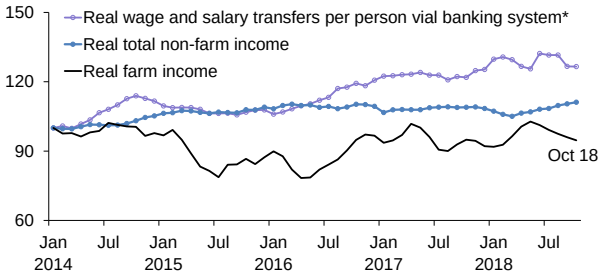
sectors still retained permanent workers but increased adoption of automation in place of temporary workers. Meanwhile, the slower growth of number of tourists had not yet affected employment in hotel business. Overall income of non-farm workers improved across almost all sectors and all income groups. Earnings of middle-income and high-income households continued to increase.

Employment of **farm households** remained stable. Farm income slightly contracted from the previous quarter but was still higher than the same period last year on account of increased outputs in almost all product categories. Some crop prices fell due to large supplies in rubber and palm oil. However, government relief measures for rice, rubber and oil palm farmers partly shored up those of purchasing power. In 2019, increasing risk of drought due to the effect of El Nino could reduce some crop production such as rice and sugar cane.

Chart 2.1 Overall purchasing power gradually improved

Household income indicators

Index, seasonally adjusted (3-month moving average)
(Jan 2014 = 100)



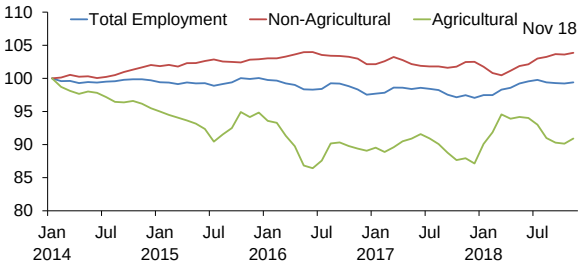
Note: *wage and salary transfer transactions are calculated from 2 databases: (1) Commercial banks reporting transactions to Bank of Thailand database which covers 90% of all retail transfer transactions and (2) Interbank Transaction Management and Exchange (ITMX) database which covers 10% of all retail transfer transactions.

Sources: Bank of Thailand, Office of Agricultural Economics, National Statistical Office, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.2 Overall employment continued to grow

Employment indicators

Index, seasonally adjusted (3-month moving average)
(Jan 2014 = 100)



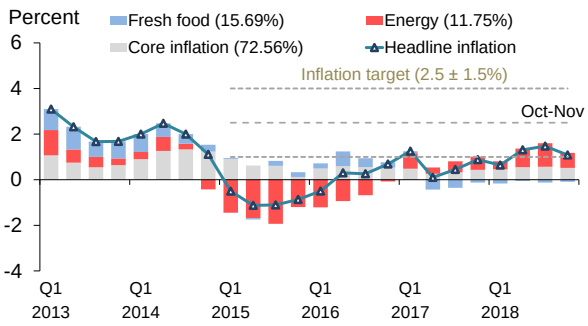
Source: National Statistical Office

Headline inflation declined mainly due to the fall in energy prices. Core inflation declined as the high base effect of excise tax increases on tobacco and alcoholic beverages was over.

Headline inflation averaged at 1.09 percent over the first two months of the fourth quarter of 2018, down from 1.47 percent in the previous quarter (Chart 2.3). The drop was due to decelerated growth of energy prices as domestic retail oil prices fell in line with global oil prices. Fresh food prices declined to a lesser degree than the previous quarter as prices of meat and eggs gradually rose after alleviation of excess supply.

Chart 2.3 Headline inflation decreased from the previous quarter mainly due to energy prices

Headline inflation and inflation target



Note: ( ) denotes share in inflation baskets
Source: Ministry of Commerce, calculations by Bank of Thailand

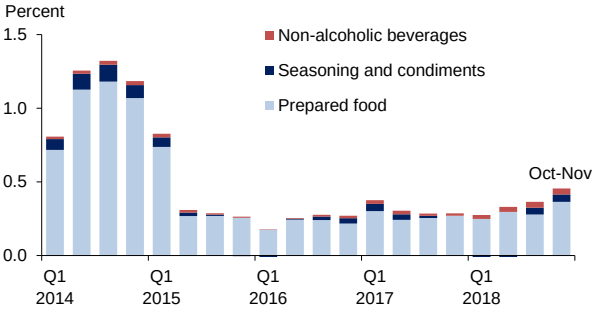
Core inflation averaged at 0.72 percent over the first two months of the fourth quarter of 2018, down from 0.78 percent in the previous quarter. Core inflation in the food category rose slightly (Chart 2.4) due to increased prices of processed foods. Core inflation in the non-food categories dropped (Chart 2.5) as the high base effect of excise tax increases on tobacco and alcoholic beverages was over. In other categories, prices rose slowly as demand pressures gradually increased. Moreover, structural factors including rising e-commerce

trends, intensified price competition, and lower production costs from higher production efficiency weighed down core inflation in non-food categories.

Short-term (one-year ahead) inflation expectations according to the survey of businesses in November 2018 stood at 2.0 percent, unchanged from the previous quarter. Inflation expectations of professional forecasters in December 2018 stood at 1.3 percent, down slightly from 1.4 percent in the previous quarter. **Long-term (five-year ahead) inflation expectations** according to the survey of professional forecasters in October 2018 stood at 1.8 percent, down from the previous survey (April 2018).

Chart 2.4 Food-in-core inflation (28% of core inflation) increased slightly mainly due to prepared food prices

Contribution* to food-in-core inflation

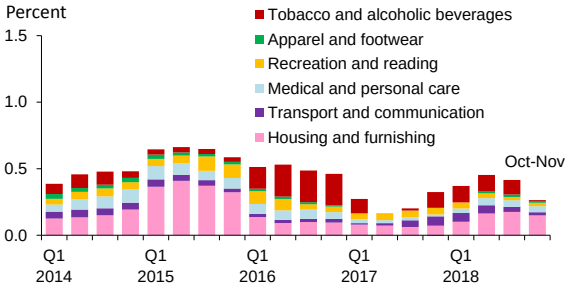


Note: *Contribution to core inflation shows decomposition of core inflation changes according to weights of each good in core inflation basket

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.5 Non-food in core inflation (72% of core inflation) decreased as the base effect of the excise tax increases on tobacco and alcoholic beverages last year waned off

Contribution* to non-food in core inflation



Note: *Contribution to core inflation shows decomposition of core inflation changes according to weights of each good in core inflation basket

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Table 2.1 Inflation

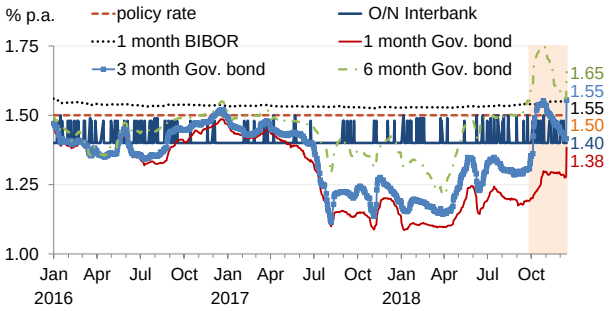
Annual percentage change	2016	2017				2018			
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Oct-Nov
Headline Consumer Price Index (Headline CPI)	0.69	1.25	0.10	0.45	0.88	0.64	1.31	1.47	1.09
Core Consumer Price Index (Core CPI)	0.73	0.66	0.47	0.49	0.61	0.61	0.76	0.78	0.72
Raw food	1.54	0.61	-2.99	-2.25	-0.80	-1.04	-0.35	-0.82	-0.58
Energy	-1.06	6.69	2.67	4.86	5.24	3.01	7.30	9.10	5.70

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Short-term money market rates remained close to the policy rate, except short-term government bond yields which were somewhat volatile mainly on account of domestic factors. Medium-term and long-term government bond yields rose in the beginning of the fourth quarter, mainly in tandem with U.S. government bond yields, but gradually fell in the latter half of November owing to domestic factors and investor concerns over external factors.

Chart 2.6 Short-term interest rates remained close to the policy rate, except short-term government bond yields which fluctuated due mainly to domestic factors

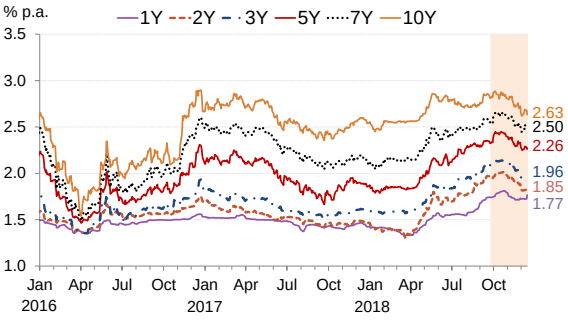
Short-term rates in financial markets



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA) (data as of 18 December 2018)

Chart 2.7 Medium-term government bond yields increased during the beginning of the quarter following US treasury yields before gradually decreasing due to increased demand by foreign investors, lower-than-expected growth outturns in Q3, and growing investor concerns over uncertainties on the external front

Government bond yields



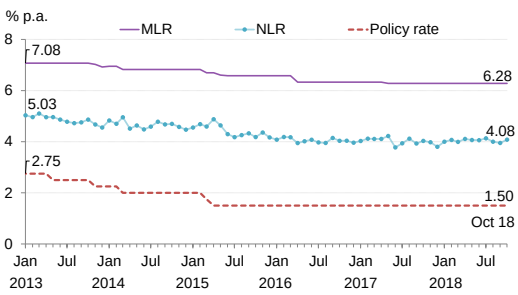
Source: Thai Bond Market Association (Thai BMA) (data as of 18 December 2018)

Most **short-term money market rates** remained close to the policy rate in the fourth quarter of 2018. **Short-term government bond yields** rose rapidly at the beginning of the quarter (Chart 2.6), partly on account of higher supply of 3-month and 6-month bonds issued by the Bank of Thailand. However, short-term government bond yields declined in November, partly on account of higher demand as a large amount of Bank of Thailand bonds would reach maturity in December. However, prior to the 8th/2018 Monetary Policy Committee meeting, short-term government bond yields rose sharply reflecting market expectations of a policy rate increase. **Medium-term and long-term government bond yields** rose in the beginning of the quarter (Chart 2.7) following the 10-year U.S. government bond yield which reached the highest point in the past seven years. The rise partly reflected market views on the opinion of several members of the Federal Open Market Committee in favor of policy rate hikes. However, in the latter half of November, yields on medium-term and long-term government bonds reduced on account of: (1) increased demand for some government bond series following the weight adjustment in the global bond index, (2) economic outturns for the third quarter of 2018 below market expectations, and (3) market concerns over the U.S. trade protectionism measures and political uncertainties in Europe, particularly Brexit negotiations and the passage of the Italian government budget bill.

Corporate bond yields slightly edged up in line with government bond yields and the slight increase in credit spread.[2/] **Cost of financing through commercial banks**, as represented in the new loan rate (NLR)[3/] also remained low, indicating financial conditions that remained accommodative (Chart 2.8).

Char 2.8 New Loan Rate (NLR) stabilized at low level

New Loan Rate



Source: Bank of Thailand (data as of October 2018)

2/ A credit spread is the difference between corporate and government bond yields with the same tenure, reflecting an assessment on corporate bond issuers' default risks.

3/ NLR is calculated based on a weighted average of interest rates for new loan contracts extended by 14 Thai commercial banks (excluding consumer loans and loans to financial intermediaries). The data covers loans of value of 20 million baht or higher for all purposes and terms and includes both secured and non-secured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

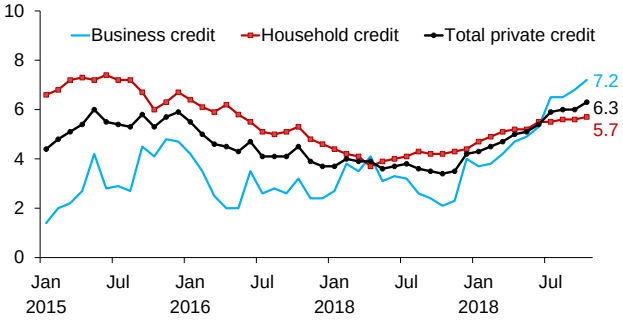
Private credit extended to both businesses and households accelerated in line with the continued economic expansion.

Private credit[4/] continued to accelerate in the third quarter and in October 2018. Private credit expanded at 6.3 percent at the end of October from the same period last year (Chart 2.9), up from a 5.4 percent growth recorded at the end of the second quarter of 2018. Business credit growth was driven by loan extended to large corporates, particularly in the commerce and service sectors, and SMEs which had relatively large credit lines such as real estate sector. Household credit continued to rise across all loan purposes in line with growing private consumption.

Chart 2.9 Private credit accelerated for both businesses and households

Growth of private credit

Percentage change from the same period last year



Note: 1) Private credit includes credit to other depository corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds
2) The data of ODCs credit outstanding to business and household sectors since January 2015 are revised following the improvement of processing system for more accurate and comprehensive ODCs credits data

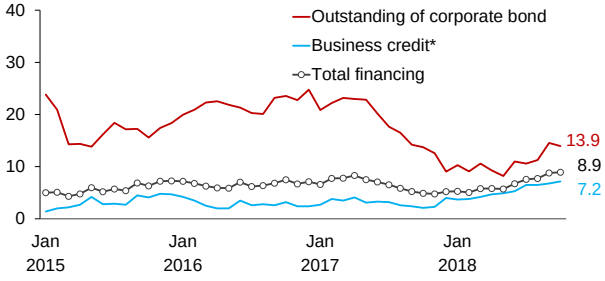
Source: Bank of Thailand

The **net issuance of corporate bonds** continued to increase in the third quarter and in October 2018 from the previous quarter, driven by funding of companies in the real estate, IT and communication, and energy sectors. In addition, the net corporate bond outstanding expanded 13.9 percent from the same period of last year (Chart 2.10). **Funding through the equity market** continued to increase in the third quarter and in October 2018, especially in the energy and utilities sectors.

Chart 2.10 Overall financing continued to expand

Growth of corporate bond outstanding and business credit

Percentage change from the same period last year



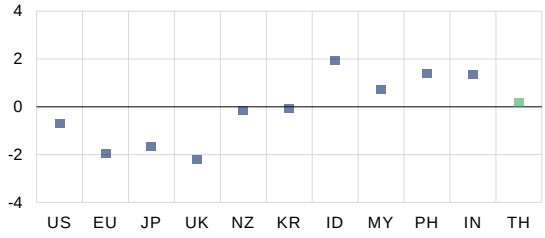
Note: *Business credit covers lending activities of Other Depository Corporation (ODCs)

Sources: Thai Bond Market Association (Thai BMA) and Bank of Thailand

Chart 2.11 Thailand's real policy rate* slightly increased but remained accommodative overall. The rate was moderate compared with other countries

Real policy rates*

Percent



Note: *Calculated from policy rate subtracted by one-year-ahead inflation expectation according to a survey by Consensus Economics (as of 10 December 2018)

Sources: Bloomberg, Consensus Economics, calculation by Bank of Thailand

4/ Outstanding credit of other depositary corporations (ODCs), namely commercial banks, specialized financial institutions, finance companies, savings cooperatives, and market mutual funds.

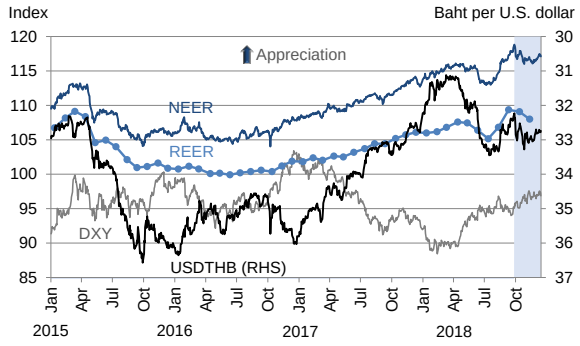
Going forward, financial conditions were expected to remain accommodative. The real policy interest rate increased slightly but still remained accommodative overall and was moderated compare with other countries (Chart 2.11). Meanwhile, costs of financing through commercial banks, as reflected in the new loan rate (NLR), would stabilize at a low level. Financial institutions[5/] were expected to tighten credit standards in the fourth quarter, particularly for loans extended to SME, to comply with the single financial account scheme. However, financial institutions would still maintain their credit standards for loans extended to households with exception of auto hire purchase loans, in which some financial institutions would somewhat loosen the credit standards. This was due to intense competition in the auto leasing market following an expiration of the first-car scheme's five-year holding period.

In the beginning of the fourth quarter of 2018, the baht depreciated against the U.S. dollar and the nominal effective exchange rate also depreciated, driven by external factors.

In the fourth quarter of 2018, **the baht depreciated against the U.S. dollar** relative to the end of the previous quarter (Chart 2.12). In particular, in October, the baht weakened against the U.S. dollar owing to market concerns over the U.S. trade protectionism measures and political uncertainties in Europe. The baht depreciation was more pronounced than regional currencies, partly due to its appreciation at a faster pace than these currencies in the previous period. However, since November, the baht strengthened slightly as the U.S. dollar weakened after concerns over the U.S. trade protectionism measures alleviated and markets expected the U.S. policy rate increases to be more gradual. Consequently on 18 December 2018, the baht closed at 32.74 to the U.S. dollar, down 1.2 percent from the end of the previous quarter.

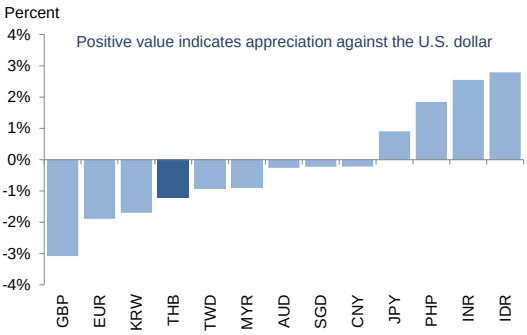
The nominal effective exchange rate (NEER) index stood at 117.11 on 18 December 2018, a 1.1 percent depreciation from the end of the previous quarter. The movement was in line with the baht depreciation against currencies of most trading partners, particularly regional currencies, as central banks in the region raised policy rates and implemented domestic and external stabilization measures to enhance sentiments of foreign investors (Chart 2.13). As the end of November 2018, the **real effective exchange rate (REER)** declined around 2.2 percent from the end of the previous quarter. Going forward, exchange rates would likely remain volatile due to uncertainties over monetary-fiscal policies of the major economies, the U.S. trade protectionism measures, geopolitical risks, particularly uncertainty over the Brexit negotiations.

Chart 2.12 The baht depreciated against the U.S. dollar as the U.S. dollar strengthened
USDTHB, NEER, DXY



Source: Bank of Thailand and Reuters (data as of 18 December 2018)

Chart 2.13 The baht depreciated against the U.S. dollar, while some regional currencies appreciated
(18 Dec 18 compared to 28 Sep 18)



Source: Bank of Thailand and Reuters (data as of 18 December 2018)

[5/] Survey of credit conditions for the third quarter of 2018 and outlook for the fourth quarter of 2018.

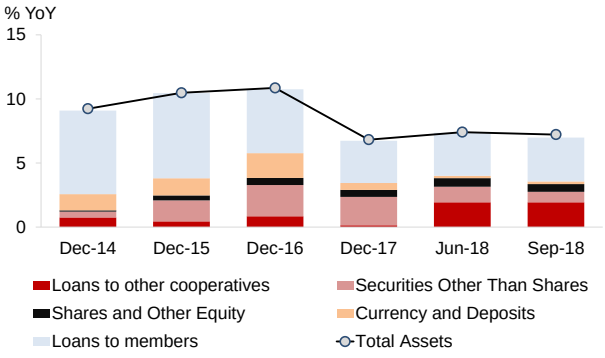
Financial stability remained sound overall. However, there remained pockets of risks that warranted monitoring. These included: (1) continued search-for-yield behavior particularly in the case of saving cooperatives and large corporates, (2) fragilities in the mortgage loan market and an oversupply of property in the real estate market, (3) elevated, albeit gradually declining, household debt and worsening debt serviceability of certain households and small enterprises, and (4) volatilities in global financial markets which could affect financing costs of the private sector.

Thailand's financial stability remained sound overall. External stability continued to be strong, as indicated by the country's sustained current account surpluses and high level of international reserves, while the external debt to GDP ratio remained low[6/]. These factors combined to cushion the Thai economy against volatilities in global financial markets. Financial institutions maintained robust financial positions, as reflected in high levels of capital buffers of commercial banks to cushion against risks stemming from deterioration in credit quality. Nevertheless, there remained pockets of risks that warranted monitoring going forward. Such risks were as follows.

(1) The continued search-for-yield behavior under the environment of persistently low interest rates could lead to underpricing of risks. Although systemic risks remained limited, there remained issues that warranted close monitoring. Such risks included, first, a search-for-higher-yield of saving cooperatives. Consequently, saving cooperatives' assets and deposits continued to expand at a high growth (Chart 2.14) despite slowing somewhat after regulatory authorities collaborated to enhance supervision standards. Meanwhile, credit risk and liquidity risk warranted monitoring owing to higher short-term borrowings by some large saving cooperatives for investment in debt instruments. In addition, increased borrowings among saving cooperatives heightened their interconnection, resulting in overall increased systemic risks in the saving cooperatives sector. Second, large corporates having greater connectedness with the financial system increasingly raised funds through bond and equities markets given a prolonged period of low interest rates. In addition, many of these companies expanded non-core business investments and outward direct investments. Furthermore, the business structure of such companies had become increasingly complicated, thus resulting in more difficult risk assessment and possibly underpricing of risk. Third, although offshore investments through foreign investment funds (FIF) had contracted in the recent period on concerns over volatilities in global financial markets, concentration risks remained as FIF investments were concentrated in five major countries.

Chart 2.14 Saving cooperatives are more inclined to search for yield behavior, especially in debt securities and loans to other cooperatives

Contribution to growth of savings cooperatives' assets



Note: Saving cooperatives were subjected to tighter regulation by government since H2/2017

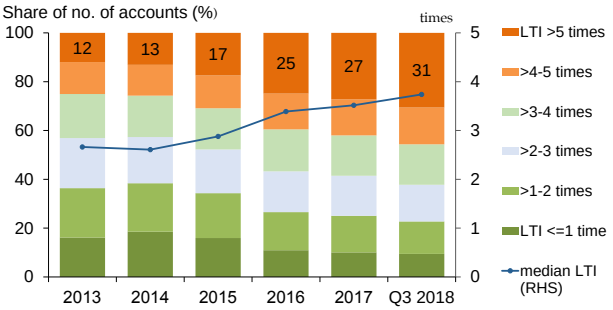
Source: Cooperative Auditing Department, calculations by Bank of Thailand

[6/] The latest data as of Q3 2018 was 35.6 percent.

(2) Signs of vulnerabilities in the mortgage loan market remained in the real estate sector. Financial institutions' competition in mortgage loan extension with a willingness to bear higher risks resulted in loosening credit standards. This was reflected in a higher median of the loan-to-income (LTI) ratio of borrowers and a rising shares of accounts with LTI ratio over five times in the third quarter of 2018 compared to the same period last year (Chart 2.15). In the period ahead, the effect of the new[7/] macroprudential me asure on the property sector and the quality of mortgage loans warranted monitoring. Prior to the effective of the regulation on April 1, 2019, demand for residences was expected to trend up owing to an acceleration of the purchase of a second or more homes. In the meantime, developers may also employ active marketing strategies to induce customers to transfer ownerships before the new regulation come into effect. **Furthermore, an oversupply in the property market still warranted monitoring.** In the previous periods, foreign demand, especially from the Chinese buyers, had played an increasing role in the condominium market. Close monitoring was, therefore, warranted as such demand may subside if the Chinese economy were to slow down.

Chart 2.15 The ratio of mortgage loan to income (Loan-to-income: LTI) and a number of accounts with LTI higher than five times in the third quarter of 2018 increased from last year

Mortgage loan-to-income ratio



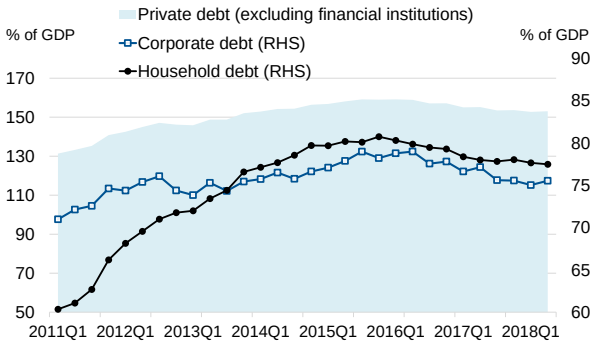
Note: Calculations from database of commercial banks' new mortgage loans

Source: Bank of Thailand

(3) Household debt remained high, albeit gradually declining, while debt serviceability of some households and small enterprises continued to deteriorate. The ratio of household debt to GDP declined slightly in the second quarter of 2018 (Chart 2.16) but remained elevated when compare with other emerging economies and had yet to show clear signs of deleveraging going forward. In particular, (1) there might be an acceleration in mortgage loans before the new regulations come into effect, and (2) credit standards on auto hire purchase loans became looser due to intensified competition among lenders. Besides, persistently low interest rates induced households to incur new debts, which could weigh on future consumption and debt serviceability and result in limited households' cushion against economic volatilities. Overall credit quality remained stable as indicated by the NPL ratio of commercial banks in the third quarter of 2018 which was stable at 2.94 percent. However, the NPL ratio in mortgage loan remained high while in auto hire purchase and credit card started to trend up. Financial positions of some small enterprises particularly those adversely affected by structural changes remained vulnerable, such as rice mills, wholesale and retail businesses. These were reflected in the sustained negative operating profit margins (OPM) and the interest coverage ratio (ICR) of

Chart 2.16 The ratio of household debt to GDP had yet to show clear sign of deleveraging

The ratio of private credit to GDP



Source: Bank of Thailand

[7/] See further at https://www.bot.or.th/Thai/PressandSpeeches/Press/News2561/n7261t.pdf

small companies[8/] which were consistent with the NPL ratio of commercial banks' SME loan portfolios which remained high at 4.7 percent.[9/]

(4) Volatilities related to capital flows heightened due to concerns over the U.S. trade protectionism measures, the outlook for the U.S. policy rate increases, and uncertainties pertaining to the Brexit negotiations. These factors could trigger yield snapbacks and affect short-term debt rollovers. Although the majority of corporate bonds issued had A or higher credit ratings, the upward trends in interest rates in global financial markets could lead to higher cost of financing through bond markets and affect financial positions of some companies that had to roll over their financing. This was particularly in the case of businesses with low or unrated credit ratings which mostly were real estate companies.

2.2 Outlook for the Thai economy

Under the Committee's assessment, the Thai economy was projected to record slower growth rates at 4.2 and 4.0 percent in 2018 and 2019, respectively, lower than the projections in the previous *Monetary Policy Report.* Key growth drivers stemming from external demand were expected to moderate as merchandise and services export growth would be lower than previously expected, consistent with slower growth in global trade volume and trading partners' economies. Meanwhile, domestic demand would remain a key growth driver, especially on the back of continued private consumption growth. The inflation projection was broadly unchanged. Headline inflation slightly edged down following declining crude oil prices.

Table 2.2 Forecast summary

Percent	2017*	2018	2019
GDP growth	3.9	4.2 (4.4)	4.0 (4.2)
Headline inflation	0.7	1.1 (1.1)	1.0 (1.1)
Core inflation	0.6	0.7 (0.7)	0.9 (0.8)

Note: * Outturn

( ) Monetary Policy Report September 2018

Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

Summary of the key forecast assumptions

- Trading partner economies were projected to exhibit slower growth throughout the forecast horizon due to lower-than-expected growth outturns in the third quarter of 2018 and a decline in domestic demand growth. Moreover, exports growth would also slow down despite having accelerated in late 2018 prior to the full implementation of the U.S. tariff measures. Exports would also expect to see some effects of the U.S. trade protectionism measures on global trade volume and trading partner economies.

[8/] As reflected by the ICR and OPM of 25th-percentile small-sized listed companies which have been in the negative territory since 2013.

[9/] The average ratio of NPL for SMEs during 2013-present was 3.9 percent.

- The federal funds rate was along the normalization path as previously projected. The Fed was expected to raise the rate four times in total in 2018 and twice more in 2019 while gradually reducing its balance sheet according to the announced plan.
- Asian currencies (excluding the Chinese yuan), throughout the forecast horizon, were revised to be weaker than the previous assessment after the outturns of Asian currencies during the fourth quarter were weaker than expected. This was a result of larger-than-expected impact from a currency crisis in some emerging market economies. Nevertheless, Asian currencies were expected to gradually appreciate in the period ahead in line with economic fundamentals of the economies in the region.
- The Dubai crude oil price was revised down throughout the forecast horizon on account of the global economic slowdown and the higher projection of crude oil supply. The supply of oil would increase after the U.S. partially lifted sanctions on Iran's oil exports to eight of Iran's major oil importers, including China, Japan, South Korea, and India, for six months.
- Farm income was revised down throughout the forecast horizon. Agricultural output, especially for in-season rice and sugarcane, was expected to decline following drought. Moreover, agricultural prices of some products, such as rubber and palm oil, was projected to fall on the back of rising global supply.
- Public spending at current prices was revised down throughout the forecast horizon mainly due to lower investment spending. Investment by central government was revised down due to delayed construction of the Bang Yai-Kanchanaburi motorway project. Moreover, delays in drafting the master plans under the National Strategy prompted government agencies to defer obligation of funds for some projects. For state-owned enterprises, investments were revised down due to the deferment of the second phase of the dual-track railway project which was under review with regard to project priority. Government consumption was projected to grow in line with the previous assessment.

Table: Summary of forecast assumptions

Annual percentage change	2017*	2018	2019
Dubai crude oil price (U.S. dollar per barrel)	53.1	69.8 (70.3)	66.0 (69.8)
Farm income (% YoY)	1.5	0.8 (2.9)	-0.4 (2.3)
Government consumption at current price (billion baht)[1/]	2,532	2,645 (2,656)	2,786 (2,786)
Public investment at current price (billion baht)[1/]	926	977 (992)	1,056 (1,082)
Fed funds rate (% at year end)	1.38	2.38 (2.38)	2.88 (3.13)
Trading partners' GDP growth (% YoY)[2/]	3.9	3.6 (3.8)	3.4 (3.5)
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)3/	155.7	153.9 (153.2)	158.7 (153.8)

Notes: [1/] Assumption includes spending on infrastructure investment plans

[2/] Weighted by each trading partner's share in Thailand's total exports

[3/] Increasing index represents depreciation, decreasing index represents appreciation

* Outturns

( ) Monetary Policy Report September 2018

The value of merchandise exports was expected to slow down due to the impact from the U.S. trade protectionism measures and the slowdown in global trade volume.

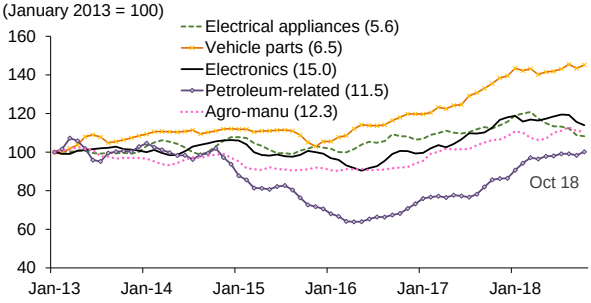
The value of merchandise exports in 2018 and 2019 was expected to record slower growth than the assessment in the previous *Monetary Policy Report*. The downward revision was due to declining export prices following a fall in crude oil prices as well as lower export volume. In September 2018, merchandise exports contracted due to a temporary factor pertaining to natural disaster in Japan, Hong Kong and the Philippines. Moreover, external demand would moderate in the period ahead as a result of (1) U.S. trade protectionism measures and the down cycle of electronic products that would weigh on global trade volume, and (2) a slowdown in external demand from trading partners consistent with economic conditions in many countries (Chart 2.17 and 2.18). However, exporters of some manufacturing industries, such as electronics parts and high-quality polyethylene resins, started to gain more benefits from redirected orders and relocation of production from China to Thailand which would support Thai exports going forward. **The value of merchandise exports was projected to expand at 7.0 and 3.8 percent in 2018 and 2019, respectively, down from the projections of 9.0 and 4.3 percent in the previous *Monetary Policy Report*.**

Under the Committee's assessment, although concerns over the U.S. trade protectionism measures alleviated in the short term, there remained uncertainties which could put pressures on international trade and investment, which would in turn affect Thai exports in both direct and indirect ways. Thus, the Committee would closely monitor developments of trade policies and negotiations, their impact on supply chains, and intensifying competition through trade diversion which could affect Thai businesses.

Chart 2.17 Merchandise exports continued to expand across various product categories, except electrical appliances to the U.S. which declined following the impacts of the U.S. trade protectionism measures on imports of solar cells and large washing machines

Value of merchandise exports, by product category

Seasonally adjusted index, 3-month moving average
(January 2013 = 100)

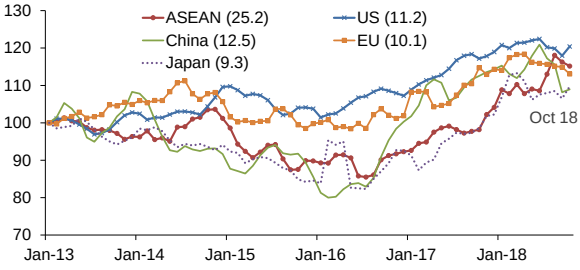


Note: Number in () denotes share to total exports in 2017
Source: Customs Department, calculations by Bank of Thailand

Chart 2.18 Merchandise exports expanded across almost all export destinations

Value of merchandise exports, by export destination

Seasonally adjusted index, 3-month moving average
(January 2013 = 100)



Note: Numbers in ( ) represent share in total exports in 2017
Source: Customs Department, calculations by Bank of Thailand

Exports of services were projected to exhibit slower growth due to a sluggish recovery in Chinese tourist figures.

Overall exports of services were projected to exhibit slower growth than previously assessed due to lower-than-expected tourism receipts resulted by fewer number of tourists and lower spending per tourist. The projected number of foreign tourists was revised down to 38.0 and 40.0 million for 2018 and 2019, respectively, from the

estimates of 38.3 and 40.6 million in the previous *Monetary Policy Report*. The downward adjustment was mainly due to Chinese tourists particularly in October 2018 when the Chinese tourist arrivals contracted 19.8 percent from the same period last year although the number of foreign tourists in the third quarter of 2018 remained close to the estimates. This reflected more severe and longer-than-expected impacts from Phuket tour boat sinking incident in early July which particularly led to a large contraction of tour groups. Such impact was indeed observed in a broader area in addition to coastal tourism destinations which were initially assessed to be the main area. On the other hand, free independent travelers (FIT) continued to exhibit robust growth. Moreover, Chinese tourists were additionally affected by China's economic slowdown and the weaker Chinese yuan. However, there were some signs of improvements in Chinese tourists after the government promptly enhanced safety standards in the tourism sector to boost confidence, put in place marketing incentives such as the free visa-on-arrival for 21 countries, and requested collaboration with the private sector to offer promotion campaigns at marketplaces. Such initiatives were expected to restore tourism growth in the period ahead.

The current account was projected to record a 33.1 billion U.S. dollars surplus in 2018, down from the estimated surplus of 35.4 billion U.S. dollars in the previous *Monetary Policy Report*. This revision was due to the downward revision of the value of merchandise and services exports. **The projection for the current account surplus in 2019 was 34.5 billion U.S. dollars, down from the estimated surplus of 36.3 U.S. billion dollars** due to the declined value of services exports.

Private consumption was projected to continue expanding.

Private consumption growth was expected to continue in almost all categories, especially durable and semi-durable goods. In particular, car sales expanded significantly due partly to the fading effect of the first-car scheme with its five-year car holding requirement. Private consumption growth would also be supported by increasingly broad-based non-farm income, continued improvements in employment in major sectors, as well as additional supports from government measures such as the state welfare cards project. However, private consumption growth was expected to moderate somewhat in the period ahead due to a number of factors: a high-base effect of durable consumption in 2018, a slowdown in farm income following the downward trend of agricultural prices, and a slowdown in non-farm income for some sectors following the outlook for merchandise and services exports. Moreover, elevated household debt would affect households to allocate part of their income for debt repayment. Greater adoption of automation in place of human labor in the production process and low inflation would lessen upward pressures on wages. As a result, average wages of Thai workers remained low and purchasing power would gradually recover in the period ahead.

Public spending was expected to moderate mainly due to investment expenditure. Meanwhile, government consumption remained close to the previous estimates.

Public spending was expected to be lower than the previous assessment due to delays in public investment by both the central government and state-owned enterprises. For the central government, investment was revised down partly due to the delayed approval of land expropriation expenses for the Bang Yai-Kanchanaburi motorway

project. Moreover, delays in drafting the master plans under the National Strategy prompted government agencies to defer obligation of funds in some projects for reviewing consistency of their budgeting plans. For state-owned enterprises, investment was revised down due to the deferment of the second phase of the dual-track rail project which was during the process of project priority review despite increased investment expenditure in 2020 according to the third phase of the Laemchabang Port Development Project by Port Authority of Thailand. Meanwhile, government consumption growth was projected close to the previous assessment.

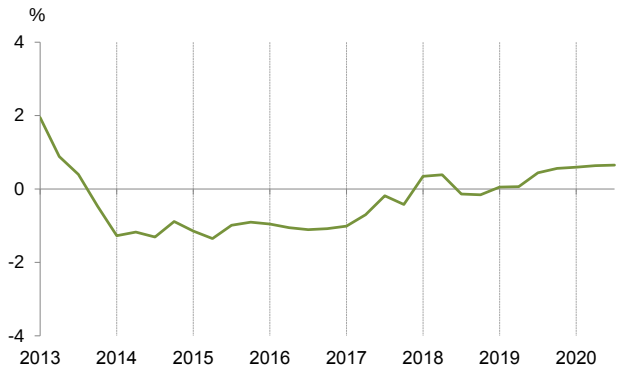
Private investment was projected to expand.

Private investment growth projections were largely unchanged, but would gradually improve mainly on the back of machinery and equipment investment. Drivers would come from upgrades to production efficiency and capacity expansion as well as the relocation of production base to Thailand. Such factors would compensate for the moderating investment trend owing to investor concerns over trade protectionism measures as well as delays in public investment. Additional factors supporting the investment outlook were (1) special measures by the Board of Investment (BOI) to boost investment and (2) the public-private partnership (PPP) in infrastructure investment projects in the Easter Economic Corridor (EEC) including the three-airport high-speed rail project. Progress of such projects would be monitored going forward.

The inflation outlook was expected to be in line with the previous assessment.

Looking ahead, headline inflation was projected to be lower than the assessment in the previous *Monetary Policy Report* due to lower crude oil prices despite some offset by higher fuel adjustment surcharge (Ft). Core inflation was expected to slightly increase due to higher bus fares announced by the Central Land Transport Control Board on 14 December 2018 and gradual improvement in demand-pull inflationary pressures, partly reflecting that output gap would gradually close in the period ahead (Chart 2.19). **The Committee thus projected headline inflation to be 1.1 and 1.0 percent for 2018 and 2019, respectively, and core inflation to be 0.7 and 0.9 percent for 2018 and 2019, respectively.**

Chart 2.19 Output gap

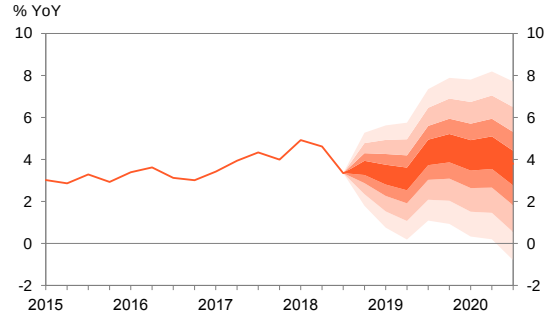


Risks to the growth and inflation projections were expected to tilt more downward.

Under the Committee's assessment, **there were increased downside risks to the baseline growth and inflation projections when compared to the assessments in the previous *Monetary Policy Report*.** These were reflected by the fan chart that tilted more downward throughout the forecast horizon (Chart 2.20) primarily due to risks on the external front as the probability of the no-deal Brexit heightened. Moreover, **the possibilities that the Thai economy would underperform the baseline projection remained.** First, impacts of the U.S. trade protectionism measures could be larger than expected. Second, Thailand's trading partner economies could expand at lower rates than anticipated in the case of

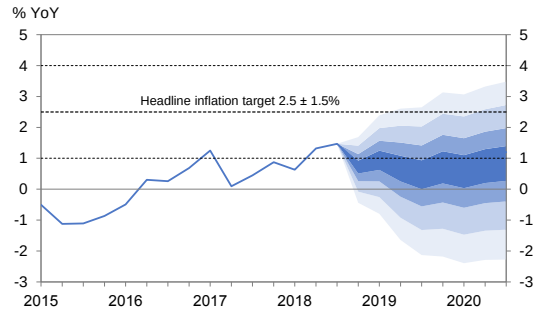
intensifying global geopolitical tensions. Third, private consumption growth could slow down due to an uneven recovery of domestic purchasing power especially for farm income. Fourth, public expenditure could be lower-than-expected owing to constraints in budget disbursement and project execution. On the other hand, **there were possibilities that the Thai economy would outperform the baseline forecast**. First, The Chinese economic slowdown could also be less severe than expected. Second, domestic spending could be higher than expected on the back of public infrastructure investment and PPP projects as well as government measures to help stimulate private consumption. Meanwhile, **risks to the forecasts of headline and core inflation were expected to tilt more downward in line** (Chart 2.21 and Chart 2.22) with increased downside risks to the growth projections and to oil prices.

Chart 2.20 Growth forecast



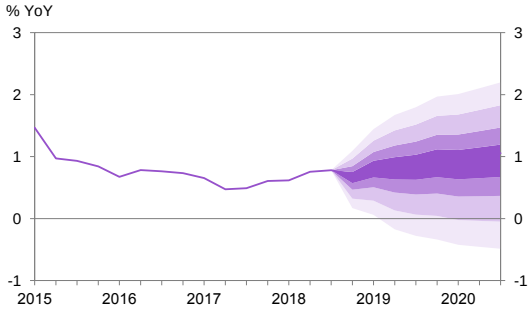
Note: Fan chart covers 90% of the probability distribution

Chart 2.21 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.22 Core inflation forecast



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2017*	2018	2019
GDP growth	3.9	4.2 (4.4)	4.0 (4.2)
Domestic demand	2.1	4.0 (3.9)	4.0 (3.8)
Private consumption	3.2	4.7 (4.2)	4.0 (3.7)
Private investment	1.7	3.6 (3.7)	4.5 (4.5)
Government consumption	0.5	2.1 (2.3)	2.6 (2.2)
Public investment	-1.2	4.6 (6.1)	6.6 (7.7)
Exports of goods and services	5.5	3.7 (5.5)	4.1 (4.1)
imports of goods and services	6.8	7.6 (7.5)	3.2 (3.3)
Current account (billion, U.S. dollars)	51.1	33.1 (35.4)	34.5 (36.3)
Value of merchandise exports	9.8	7.0 (9.0)	3.8 (4.3)
Value of merchandise imports	13.2	15.3 (16.9)	3.8 (5.6)
Number of foreign tourists (million person)	35.4	38.0 (38.3)	40.0 (40.6)

Note: *Outturns

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3. Monetary Policy Decision

The Committee assessed that the Thai economy would continue to expand at a level consistent with its potential. Headline and core inflation were expected to be broadly in line with the previous assessment. Financial stability remained sound overall but there remained a need to monitor risks that might pose vulnerabilities to the financial system in the future.

The Committee placed great emphasis on the strength and continuation of economic growth, development of headline inflation, and preserving financial stability. The Committee assessed that the economy was expected to continue gaining traction driven by domestic demand momentum, despite the slowdown in external demand. Nevertheless, the possibility that the Thai economy would expand slower than the baseline projection increased, due mainly to risks on the external front. Inflation would gradually rise. Meanwhile, there remained a need to monitor pockets of risks to financial stability arising from a prolonged low interest rate environment.

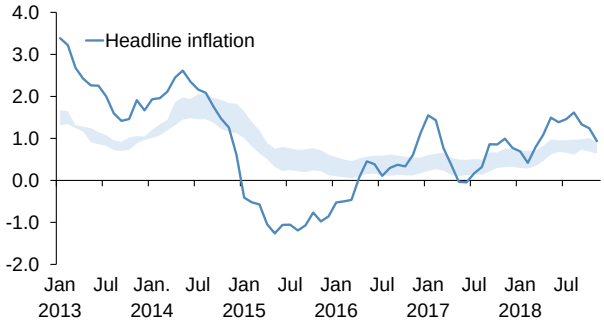
1. Economic growth. The Thai economy was expected to continue expanding at a rate consistent with its potential. In particular, private consumption expanded on account of improvements in the labor market for both employment and income, as well as government measures aiming to enhance household purchasing power. However, a relatively high level of household debt remained a factor weighing on private consumption. Furthermore, private investment would likely expand, partly benefited from investment aiming to improve efficiency and expand production capacity and the relocation of production base of certain industries to Thailand. Moreover, there were also supporting factors from the EEC and the PPP projects. However, public expenditure would contribute less as an economic driver due to delays in some investment projects and deferring of obligation of funds for some projects by government agencies to await for adjusting their work plans to align with the delayed drafting of master plans under the National Strategy. Government consumption outlook was broadly unchanged. However, external demand momentum would slow down as exports of some products were affected by trade protectionism measures, as well as a consequence of a down cycle of electronic industry and a slowdown of trading partner economies, especially China which had great linkages with the Thai economy. Exports of services would likely slow down following a declining number of Chinese tourists although there were some signs of slight improvement in Chinese tourist figures. Nevertheless, risks that the Thai economy would underperform the baseline projection increased due mainly to risks of the no-deal Brexit.

2. Inflation developments. Annual average of headline inflation for 2018 and 2019 were projected to be within target, but there was a possibility that headline inflation might fall below the baseline projection in line with risks to growth forecasts and oil prices that might be lower than expected. Core inflation was expected to somewhat increase, partly due to upward adjustment of bus fares, as well as gradually rising demand-pull inflationary pressures, as reflected in various indicators including trimmed mean and principal component indicators. (Chart 3.1) This was partly resulted from structural changes which contributed to more persistent inflation than in the past. Meanwhile, **inflation expectations** according to surveys from businesses and professional forecasters were largely unchanged from the previous quarter. (Chart 3.2)

Chart 3.1 Demand-pull inflationary pressures gradually increased

Underlying inflation indicators

Percent change from same period last year



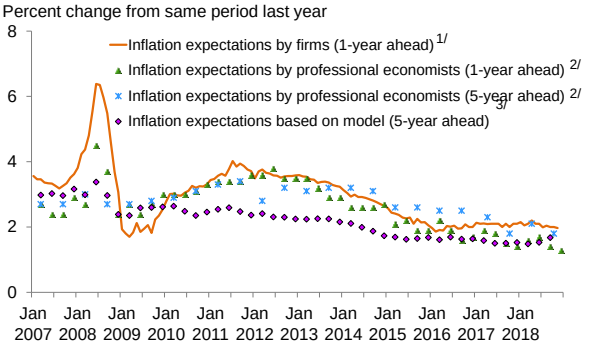
Note: The shedded area shows the highest and lowest outcomes among different underlying inflation indicators. The indicators include 1. **Trimmed mean** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute to price movements across categories of goods and services) and 3. **Core inflation excluding rent and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 3.2 Inflation expectations remained broadly unchanged from the previous quarter

The public inflation expectations

Percent change from same period last year



Source: 1/ Business Sentiment Index (BSI) Survey of Bank of Thailand
2/ Asia Pacific Consensus Forecast
3/ Macro-finance term structure model calculated from bond and macroeconomic data

3. The build-up of risks to financial stability. The Committee viewed that financial stability remained sound overall, but there was a need to monitor certain pockets of risks that had yet to show clear signs of improvement and might pose vulnerabilities to the financial system in the future. These risks included **(1) search-for-yield behavior which could lead to greater underpricing of risks given a prolonged low interest rate environment.** This was seen, for instance, in a continued expansion in both deposit and investment assets of saving cooperatives which continued to provide high returns to members, despite some decelerated asset growth after an enhancement of regulations on saving cooperatives. In addition, greater linkages among saving cooperatives, which resulted from increased lending activities among each other, heightened systemic risks in the cooperative system. Moreover, large corporates would likely seek funding through the bond market more for investment in core and non-core businesses, as well as overseas businesses. This could lead to underpricing of risks given low interest rate environment. More complexity of these business structures caused risk assessment to be more challenging. Another risk was **(2) a build-up of vulnerabilities in the mortgage loan market in the real estate sector.** The intensified competition in the mortgage loan market resulted in the financial institutions willing to accept higher risks and looser credit standards. There remained a need to monitor the impact of the revision in macroprudential measure on mortgage loans on the property market and mortgage loan quality, especially the adjustments of the market during the period prior to the effective date. Moreover, the oversupply in the property market remained an issue to be followed up. In particular, external demand, especially from China which played an increasing role in the condominium market, might decline if the Chinese economy slowed down. The other risk involved **(3) the relatively high ratio of household debt to GDP despite a slow decline** following an expansion in all types of consumer loans, especially auto leasing. Moreover, a low interest rate induced household to borrow more and save less. This would affect the amount of future savings, especially as Thailand was entering aging society.

Amid increasing interconnectedness between the real economic and financial sectors, the monetary policy conduct aiming solely to pursue price stability would not be sufficient. It was deemed necessary to place emphasis on preserving financial stability as well. In this light,

the Committee viewed that **a complementary use of policy instruments, both policy rate and macroprudential measures, would be necessary to effectively preserve financial stability.** Macroprudential measures could not be employed in place of appropriate monetary policy, as such measures were rather complementary measures to address specific pockets of risks. (Box: *Setting appropriate framework and objectives of Thailand's monetary policy amid challenges under changing economic and financial context*)

Most Committee members viewed that the need for extra accommodative monetary policy as in the past had reduced and thus voted to raise the policy rate to 1.75 percent at the meeting on December 19, 2018 to curb financial stability risks and to start building policy space whenever appropriate.

The Committee voted 4 to 3 to maintain the policy rate at 1.50 percent at the meeting on November 14, 2018. The Committee viewed that an accommodative monetary policy helped support economic growth and was in line with the inflation target. Most members thus saw a need to keep the policy rate unchanged. Three members, however, viewed that the continuation of economic growth was sufficiently robust and that prolonged monetary accommodation induced households and businesses to underestimate potential changes in financial conditions. Thus, they voted to raise the policy rate to curb financial stability risks that could affect the sustainability of economic growth over the longer term and to start building policy space. Meanwhile, the Committee viewed that should the economic expansion continue and inflation move in line with target, the need for currently extra accommodative monetary policy would begin to decline and the importance to have policy space to cushion against possible risks to growth in the future would be more prominent.

The Committee voted 5 to 2 to raise the policy rate by 0.25 percentage point to 1.75 percent at the meeting on December 19, 2018. The Committee viewed that the economy expanded at the level consistent with its potential and headline inflation moved in line with target. Most members viewed that the continuation of economic growth was sufficiently robust and the need for extra accommodative monetary policy as in the past had reduced. Thus, the policy rate should be increased to curb financial stability risks and to build policy space whenever appropriate. The policy rate at 1.75 percent was deemed to remain conducive to the continuation of economic growth. (Box: *The need to hike the policy rate at the MPC meeting 8th/2018*.) Two members viewed that risks and uncertainties on the external front heightened and could affect growth outlook of the Thai economy in the period ahead, and thus saw a need to assess the clarity of such impacts on the sustainability of domestic growth momentum for some time. In addition, the implemented macroprudential measures had addressed certain risks in the financial stability to some extent.

Looking ahead, the Thai economy was projected to continue to gain traction despite the slowdown in external demand. The Committee viewed that accommodative monetary policy would remain appropriate in the period ahead and that the policy rate increase would be gradual and not in a continuous manner as in the past. The Committee would closely monitor developments of economic growth, inflation, and financial stability, together with associated risks, in deliberating appropriate monetary policy in the period ahead.

Setting appropriate framework and objectives of Thailand's monetary policy amid challenges under changing economic and financial context.

Thailand has adopted the monetary policy framework with flexible inflation targeting (FIT) as the monetary policy objective since 2000 mainly to pursue price stability. This has helped stabilize Thailand's inflation not to move above the target range. However, rapid change in structures of the global and Thai economies in the past decade resulted in downward trend of inflation dynamics. Moreover, financial stability has become increasingly important following the global financial crisis in 2008 stemmed from the build-up of vulnerabilities in the U.S. financial system. **Given changing inflation dynamics and growing importance of financial stability, most inflation targeting central banks are now facing greater challenges and constraints in the conduct of monetary policy.**

Key challenges in the conduct of Thailand's monetary policy

1) Effectiveness of monetary policy in pursuing price stability reduced because (1) *Thailand's inflation was increasingly influenced by external factors*, especially by prices of commodities in the global market and greater interconnections in trade and financial channels between each country, and (2) *Thailand's inflation dynamics were expected to be lower than in the past* due to cyclical factors including rising agricultural production given favorable weather conditions, which pressured on prices. Structural factors including technological development also helped reduce production costs. For example, shale oil technology led to lower crude oil prices. Moreover, the rise of global value chain as a consequence of globalization in an attempt to seek raw material sources also helped reduce costs. An expansion of e-commerce helped reduce operational costs of businesses and led to intensifying price competition. In addition, Thailand's aging society could restrain consumption growth in the period ahead.

2) The role of monetary policy was extended to preserve financial stability. Financial sector has become increasingly connected to the real sector through new alternative forms of savings and investment in more developed financial market. This might lead to build-up of vulnerabilities in the financial system more easily. The conduct of monetary policy aiming to preserve financial stability would have to additionally address financial stability issues, rather than mainly pursuing price stability. This would help prevent a crisis as took place in the U.S. in 2008. Moreover, most studies on the relationship between monetary policy and financial stability suggested that an extra and prolonged accommodative monetary policy could pose risks in the financial system and increase the likelihood of financial crisis through risk-taking channel, prompting central banks worldwide to start placing more emphasis on preserving financial stability.

Appropriateness of Thailand's current monetary policy framework

Challenges mentioned above prompted central banks worldwide to investigate on other monetary policy frameworks that could be applied. Such frameworks are, for example, nominal GDP-level targeting, price-level targeting, and temporary price-level targeting at zero lower bound.[10/] The studies also include finding ways to improve the current monetary policy framework to suit the current economic context. However, some frameworks are subject to practical limitations such as lag and low frequency of GDP data which could result in monetary policy being less timely and was not helpful in addressing financial stability**. In the case of Thailand, the FIT framework remains appropriate as flexible monetary policy framework has facilitated the Bank of Thailand (BOT) in maintaining economic and financial stability of the country, while also pursuing price stability.** The BOT's currently adopted FIT framework sets inflation as the medium-term target with a band which allows inflation to fluctuate around the

[10/] Ben S. Bernanke (2017), "Temporary Price-level Targeting: An Alternative Framework for Monetary Policy", Brookings Institution.

mid point in the short run. The current framework thus allows monetary policy consideration to place emphasis on other dimensions of stability in conjunction with price stability.

The role of monetary policy when its influences on inflation decreased

Although the effectiveness of monetary policy on inflation declines during low inflation period, price stability remains a key objective of monetary policy conduct. This is because (1) Too-high or too-low inflation would affect saving and investment plans and would in turn impact economic activities and household purchasing power. On the other hand, stable inflation would help anchor public's inflation expectations, as well as help the public to plan for future economic activities more easily. (2) Pursuing price stability is consistent with basic principles of monetary policy, long-run money neutrality, in which monetary policy in the long term would affect solely on inflation but not on economic growth at potential level. Therefore, setting inflation as the monetary policy target remains appropriate and maintaining stable inflation remains an important monetary policy objective.

Nonetheless, Thailand should consider setting the inflation target to be consistent with lower inflation dynamics than in the past. An appropriate inflation target should not be too high or too low. Otherwise, the effectiveness of monetary policy could be affected. Moreover, inflation target should be flexible to allow the BOT to simultaneously maintain other stability objectives altogether.

(1) *Appropriateness of the current inflation target*. The Monetary Policy Committee (MPC) considered benefits and costs of setting inflation target at different levels, as well as their impacts on overall economic and financial stability. The MPC viewed that the plan to lower inflation target from the current level at an appropriate timing would help ensure consistency of the target to Thailand's inflation dynamics under the new context, where inflation would potentially be lower than in the past due mainly to structural factors. This would help reduce the possibility of monetary policy being excessively accommodative and reduce the possibility of inflation being lower than the target which is incompatible with the current context. In recent periods, many central banks, including the Bank of Korea and Norges Bank, lowered their inflation targets to be consistent with changing inflation dynamics driven by structural factors.

Nevertheless, **considering short- and long-run factors, the aforementioned plan on reducing the inflation target might not be applicable in 2019**. In the short run, impacts on the economic growth must be considered as public might hold up their spending if they expected a significantly less accommodative monetary policy stance in the period ahead following the inflation target reduction. This might cause a fall in inflation expectation. In the long run, monetary policy space as a buffer against crisis in the future could decrease. This was because the inflation target reduction would lower the interest rate upcycle than ever before, following a decline in medium-term inflation expectation. Thus, maintaining an annual average inflation target at 2.5±1.5 percent remained appropriate in the medium term and for 2019, and would help facilitate economic growth to be consistent with Thailand's growth potential. Meanwhile, the tolerance band of the inflation target at 1.5 percent also remained appropriate to cushion against volatilities that might cause future inflation to deviate from the mid point of the target in the short run. For consideration of the inflation target in the future, the BOT would conduct further studies on structural changes which might have impacts on inflation dynamics and inflation outlook going forward, and would assess the benefits and costs. This would help setting monetary policy target in the period ahead to be appropriate and enhance effectiveness of monetary policy conduct.

(2) *Plans on changing appropriate inflation target under the FIT framework.* The setting of inflation target should be flexible enough to allow the MPC to place emphasis on other aspects of stability in conjunction with price stability. Alternative settings included setting only the upper bound of the target, setting the target range, and reconsidering the role of monetary policy in price stability

over medium term, as seen among other central banks (Table 1). This would allow monetary policy to put greater considerations on factors related to financial stability in a more flexible manner.

Table 1 Duration setting of inflation target among various central banks

Duration setting of inflation target		Country
Medium-term target	not specified	Australia, Canada[1/], Czech Republic, New Zealand, Sweden, Norway, Armenia, Hungary, Mexico, Poland, Romania, Serbia, Colombia[2/], Thailand[2/]
	2 - 3 years	South Korea (3 years), Israel (2 years), the Philippines (3 years), Turkey (3 years), Chile (2 years), Ghana (1-1.5 years)
Short-term target (1 year)		Brazil, Guatemala
At all times		The United Kingdom, Iceland Peru, South Africa

Note: Blue indicates developed economies

[1/] In general, the duration of the target is set at 6-8 quarters, but can be modified depending on economic conditions.

[2/] The law requires an annual target setting, but in practice, medium-term target is also communicated.

Source: Gill Hammond (2012), "State of the Art of Inflation Targeting", Centre for Central Banking Studies Handbook No. 29 and compilations from central banks' websites

The role of monetary policy in preserving financial stability

The BOT should enhance the FIT framework to strengthen effectiveness and cushion against various challenges in the period ahead. In particular, financial stability should be incorporated into the process of monetary policy decision in a systemic way, including development of more in-depth and comprehensive indicators in conjunction with the use of macroprudential measures to preserve financial stability in accordance with Thailand's changing economic and financial context. The BOT has recently undertaken the following processes:[11/]

(1) Developing a financial cycle indicator to support monetary policy deliberation. If business cycle and financial cycle move in the same direction, monetary policy could preserve macroeconomic stability, price stability, and financial stability altogether at once. However, if both cycles move in different directions, monetary policy should consider trade-offs between preserving economic and inflation stability, and preserving financial stability.

(2) Developing financial stability indicators. Considering only the financial cycle indicator might not be sufficient for assessing financial stability risks, as it only takes into account financing in the financial market and activities in the real estate sector. The build-up of other vulnerabilities in the financial system should also be considered. Such vulnerabilities include underpricing of risks and risks from regulatory gaps. Developing indicators to monitor various types of risks in the financial system would help indicate the possibility of a financial crisis in the future

[11/] Warapong Wongwachara, Bovonvich Jindarak, Nuwat Nookhwun, Sophon Tunyavetchakit, and Chutipha Klungjaturavet (2018), "Integrating Monetary Policy and Financial Stability: A New Framework". Research presented at the Bank of Thailand's academic symposium in 2018.

and would facilitate a prompt response of monetary policy. In this light, financial stability dashboard was developed to monitor risks in various aspects and to be used in conjunction with analysis of the relationship between business and financial cycle. The dashboard would help in assessing pockets of risks and deciding the appropriate monetary policy tools for different types of risks.

(3) Developing macroprudential measures, as a complementary tool to support the policy rate tool, which would help address issues in a more effective and targeted manner. The BOT collaborated with other regulatory agencies in the financial sector in preparing for the use of macroprudential measures to ensure more effective mechanism in regulating and monitoring risks in the financial system. Looking ahead, institutional arrangement of Thailand's financial regulatory agencies should be set up in a more systematic manner regarding data coordination, policy tools, and implementation of regulations and regulatory guidelines. This would help enhance preserving financial stability of the country.

(4) Conducting monetary policy with greater consideration on financial stability. In case of systemic risks, which might become widespread and affect economic and financial system, or when economic and financial imbalances might dampen economic growth and inflation in the period ahead, the MPC might tighten monetary policy stance in the-lean-against-the-wind manner to cope with imbalances in the financial system. Although this might slow down inflation and economic expansion in the short run, it would support sustainable economic growth in the long run. Moreover, the MPC might deploy the policy rate tool in conjunction with macroprudential measures to enhance effectiveness of monetary policy as a preventive tool against financial stability risks from worsening economic and inflation stability. This was because macroprudential measures alone might not be sufficient in preserving financial stability overall as risks could migrate to a less regulated part of financial sector, while policy rate was able to deal with all of the cracks.[12/]

Development of monetary policy communication strategy to improve public understanding

Transparent communication regarding rationales of the decisions taken and monetary policy stance would help business sector and the public adjust themselves and plan their economic activities in advanced. The MPC thus communicated factors influencing the monetary policy decisions under the FIT framework, as well as roles and constraints of monetary policy in preserving various types of stability. In particular, **there was a need to place more emphasis on financial stability, while not abandoning economic growth and price stability objectives in the medium term.**

In recent periods, the MPC communicated through conventional channels such as *the edited minutes,* which focused on issues that the MPC discussed regarding economic and financial assessment, as well as strategic issues which would have a significant impact on monetary policy conduct in the period ahead. Another example included *the monetary policy report,* which published data supporting the MPC decision and rationales behind monetary policy decision in each quarter. The report also presented insightful analyses to enhance public understandings on monetary policy conduct. In addition, the report would also show the MPC's will in implementing monetary policy to preserve financial stability and to deepen public understandings on principles of monetary policy decisions. This would help improve public expectations on monetary policy outlook going forward.

[12/] Jeremy C. Stein (2013), "Overheating in Credit Markets: Origins, Measurement, and Policy Responses". Speech given at the "Restoring Household Financial Stability after the Great Recession: Why Household Balance Sheets Matter" research symposium sponsored by the Federal Reserve Bank of St. Louis, St. Louis, Missouri

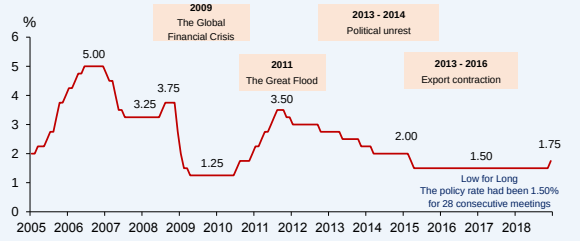
Looking ahead, the MPC would gradually utilize more communication channels, especially through online media. This would promote public access to information regarding economic and financial assessment, monetary policy conduct, and trade-offs between different objectives in making monetary policy decisions. Such communications would also conveniently enhance understandings on the monetary policy outlook in the period ahead, allowing businesses and the public to adjust themselves in a timely manner and forming appropriate expectations on monetary policy direction according to the MPC's communication strategy to the public at each meeting.

The need to hike the policy rate at the MPC meeting 8th/2018.

Accommodative monetary policy had started since November 2011 to support economic recovery from the Great Flood. During three to four years later, the MPC then steadily reduced the policy rate when global trade slowdown hit the economy by exports contraction. Meanwhile, domestic demand recovered quite slowly following weak fiscal stimulus as a result of domestic political unrest. Moreover, the MPC had voted to maintain the policy rate at 1.5 percent from April 2015 until December 2018. Since 2011, the seven-year period was considered the longest do wncycle of the policy rate. (Chart 1)

Chart 1 Monetary policy helped accommodate the economy over the periods that were affected by various factors. The policy rate remained low for a long period of the latest down cycle.

Thailand's policy rate



%	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018E	2019E	2008 - 2017
GDP	4.2	5.0	5.4	1.7	-0.7	7.5	0.8	7.2	2.7	1.0	3.0	3.3	3.9	4.2	4.0	3.0
Headline inflation	4.5	4.6	2.2	5.5	-0.9	3.3	3.8	3.0	2.2	1.9	-0.9	0.2	0.7	1.1	1.0	1.9
Value of exports	15.2	17.0	18.2	15.9	-13.4	27.1	14.8	2.8	-0.1	-0.4	-5.6	0.1	9.8	7.0	3.8	5.1

Note: E = denotes estimation by the Bank of Thailand

Chart 2 The need for extra accommodative monetary policy reduced compared with the past events.

Economic growth, headline inflation, and policy rate

MPC meeting	GDP	Headline inflation	Policy rate
3rd/2009 The Global Financial Crisis	-0.7%	-0.9%	1.25%
3rd/2015 After political unrest and export contraction	3.0%	-0.9%	1.50%
8th/2018 Present (19 Dec 2018) The economy grew in line with potential level, and inflation was within target	4.2%	1.1%	1.75%

Keeping the policy rate low at 1.50 percent throughout 28 MPC meetings contributed to continuing improvements of the Thai economy. In 2017, the economy grew as high as 3.9 percent and was expected to gain further traction at 4.2 percent in 2018 and 4.0 percent in 2019. Such growth rate was consistent with the potential level and above the average growth over the past 10 years at 3.0 percent. (Chart 1) Moreover, headline inflation was expected to gradually rise and move in line with the target in the period ahead. **The MPC thus assessed that the need to maintain the policy rate at an extra accommodative level started to reduce, especially when compared to the past situations.** (Chart 2)

However, a prolonged period of low interest rate led to a buildup of vulnerabilities in the financial system. This prompted households to continue leveraging, while discouraged savings. This would have greater impacts as Thailand is entering aging society. Moreover, the public and businesses would seek alternatives to save and invest for higher returns, possibly leading to underpricing of risks. This would easily cause imbalances in the economic and financial system and could in turn affect sustainability of long-term economic growth.

The MPC assessed that an increase in the policy rate at this meeting would not be an obstacle to economic growth and inflation outlook in the period ahead, but would help reduce risks to the financial stability, which would in turn be beneficial for the sustainability of economic growth. The MPC viewed that the use of monetary policy in conjunction with macroprudential measures that had already been implemented to a certain extent, would help preserving financial stability more effectively. In addition, an increase in the policy rate, whenever appropriate, would help build policy space under economic and financial outlook where there were high uncertainties in the period ahead, especially on the external front. Nevertheless, the MPC viewed that an increase in the policy rate would be gradual and might not be in a continuous manner as in the past. The assessments would be data dependent in order to implement an appropriate course of monetary policy in the period ahead.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2016	2017	2017		2018		
			Q3	Q4	Q1	Q2	Q3
GDP growth	3.3	3.9	4.3	4.0	4.9	4.6	3.3
Production							
Agriculture	-2.5	6.2	9.7	-1.3	6.5	10.2	4.3
Non-agriculture	3.8	3.7	4.0	4.7	4.8	4.1	3.3
Manufacturing	2.3	2.6	4.2	3.4	3.8	3.2	1.6
Construction	8.6	-2.3	-1.6	-5.3	1.2	2.0	4.7
Wholesales and retail trade	5.3	6.3	6.4	6.9	7.0	7.3	7.2
Hotels and restaurants	9.9	8.5	6.9	15.3	12.8	9.4	6.5
Transport, storage, and communication	4.1	7.3	7.4	8.8	7.5	6.8	6.2
Financial intermediation	6.5	4.8	4.6	3.6	3.6	4.6	3.0
Real estate, renting, and business activities	3.2	4.6	4.7	5.8	4.9	3.5	4.8
Expenditure							
Domestic demand	2.8	2.1	2.5	2.0	3.3	3.8	4.2
Private consumption	3.0	3.2	3.4	3.4	3.7	4.5	5.0
Private investment	0.5	1.7	2.5	2.4	3.1	3.2	3.9
Government consumption	2.2	0.5	1.8	0.2	1.9	2.0	2.1
Public investment	9.5	-1.2	-1.6	-6.0	4.0	4.9	4.2
Imports of goods and services	-1.0	6.8	6.5	7.5	8.7	8.3	10.7
imports of goods	-2.3	8.5	9.2	8.3	9.3	7.2	9.4
imports of services	4.6	-0.3	-5.0	4.0	6.2	13.3	16.9
Exports of goods and services	2.8	5.5	6.9	7.4	6.0	6.8	-0.1
exports of goods	0.3	5.6	8.2	6.6	4.7	7.4	-0.2
exports of services	11.5	5.1	2.6	9.9	9.5	4.9	0.2
Trade balance (billion, U.S. dollars)	36.5	34.2	10.6	7.0	6.6	5.8	3.4
Current account (billion, U.S. dollars)	48.2	50.2	14.9	12.4	15.1	6.7	4.2
Financial account (billion, U.S. dollars)	-20.8	-12.4	2.7	-5.3	-3.7	-5.1	-1.4
International reserves (billion, U.S. dollars)	171.9	202.6	199.3	202.6	215.6	206.8	204.5
Unemployment rate (%)	1.0	1.2	1.2	1.1	1.2	1.1	1.0
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	1.2	1.3	1.2	1.0	1.0

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2016	2017	2017		2018				
			Q3	Q4	Q1	Q2	Q3	Oct	Nov
1. Financial market sector									
Bond market									
Bond spread (10 years - 2 years)	0.6	1.1	1.0	1.0	1.1	1.1	0.9	0.9	0.8
Equity market									
SET index (end of period)	1,542.9	1,753.7	1,673.2	1,753.7	1,776.3	1,595.6	1,756.4	1,669.1	1,641.8
Actual volatility of SET index[1/]	14.2	6.5	5.8	7.9	9.4	12.3	11.6	16.9	10.4
Price to Earnings ratio (P/E ratio) (times)	18.6	19.1	17.9	19.1	18.3	16.2	17.3	16.5	15.6
Exchange rate market									
Actual volatility of Thai baht (%annualized)[2/]	4.4	3.3	2.9	2.8	4.6	4.5	4.5	5.3	5.2
Nominal Effective Exchange Rate (NEER)	106.2	110.6	111.2	112.9	114.8	115.2	115.2	117.3	116.6
Real Effective Exchange Rate (REER)	100.6	103.6	104.2	105.8	106.3	107.1	107.1	109.1	108.0
2. Financial institution sector[3/]									
Minimum Lending Rate (MLR)[4/]	6.33	6.28	6.28	6.28	6.28	6.28	6.28	6.28	6.28
12-month fixed deposit rate[4/]	1.37	1.37	1.37	1.37	1.37	1.37	1.37	1.37	1.37
Capital adequacy									
Capital funds / Risk-weighted asset (%)	18.0	18.2	18.5	18.2	18.1	17.9	18.4	18.4	n.a.
Earning and profitability									
Net profit (billion, Thai baht)	198.5	187.1	46.5	40.6	50.2	56.4	51.3	n.a.	n.a.
Return on assets (ROA) (times)	1.2	1.1	1.1	1.1	1.11	1.2	1.2	n.a.	n.a.
Liquidity									
Loan to Deposit and B/E (%)	96.3	96.1	96.4	96.1	95.0	96.8	98.2	97.2	n.a.
3. Household sector									
Household debt to GDP (%)	79.3	78.1	77.8	78.1	77.7	77.5	n.a.	n.a.	n.a.
Financial assets to debt (times)	2.7	2.6	2.6	2.6	2.7	2.6	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Consumer loans	2.7	2.7	2.7	2.7	2.8	2.7	2.7	n.a.	n.a.
Housing loans	2.9	3.2	3.3	3.2	3.4	3.4	3.4	n.a.	n.a.
Auto leasing	1.8	1.6	1.6	1.6	1.5	1.5	1.6	n.a.	n.a.
Credit cards	3.7	2.6	2.8	2.6	3.2	2.4	2.5	n.a.	n.a.
Other personal loans	2.9	2.5	2.7	2.5	2.7	2.5	2.5	n.a.	n.a.
4. Non-financial corporate sector[5/]									
Operating profit margin (OPM) (%)	8.2	8.0	8.5	7.9	7.9	7.8	8.0	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.7	0.7	0.8	n.a.	n.a.
Interest coverage ratio (ICR) (times)	6.4	6.5	6.4	7.8	7.1	6.5	6.5	n.a.	n.a.
Current ratio (times)	1.6	1.7	1.6	1.7	1.7	1.7	1.6	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Large businesses	1.5	1.8	1.7	1.8	1.7	1.7	1.5	n.a.	n.a.
SMEs	4.3	4.4	4.6	4.4	4.5	4.5	4.7	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 5 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2016	2017	2017		2018				
			Q3	Q4	Q1	Q2	Q3	Oct	Nov
5. Real estate sector									
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)									
Total	61,722	62,665	16,859	18,476	12,856	15,230	17,919	4,232	n.a.
Single-detached and semi-detached houses	13,409	13,907	3,774	3,787	3,162	3,365	3,794	904	n.a.
Townhouses and commercial buildings	20,187	20,536	5,604	5,670	4,248	4,922	5,955	1,355	n.a.
Condominiums	28,126	28,222	7,481	9,019	5,446	6,943	8,170	1,973	n.a.
Number of new housing units launched for sale (Bangkok and Vicinity) (units)									
Total	110,575	114,477	35,434	24,863	26,243	19,627	40,656	13,412	n.a.
Single-detached and semi-detached houses	19,433	14,280	4,874	3,608	3,797	2,790	6,143	838	n.a.
Townhouses and commercial buildings	32,792	36,571	9,831	7,892	6,435	6,522	8,562	5,654	n.a.
Condominiums	58,350	63,626	20,729	13,363	16,011	10,315	25,951	6,920	n.a.
Housing price index (2009 = 100)									
Single-detached houses (including land)	130.8	130.9	131.6	133.9	137.7	137.8	139.8	140.0	n.a.
Townhouses (including land)	137.6	141.2	142.6	143.7	145.9	149.5	151.7	151.6	n.a.
Condominiums	165.3	169.7	169.8	175.4	179.8	176.7	180.4	181.3	n.a.
Land	171.2	171.7	172.9	178.3	175.7	177.2	172.7	171.1	n.a.
6. Fiscal sector									
Public debt to GDP (%)	40.8	41.2	41.9	41.2	41.2	41.0	41.7	41.7	n.a.
7. External sector									
Current account balance to GDP (%)[6/]	11.7	11.0	12.9	10.1	11.7	5.3	3.4	n.a.	n.a.
External debt to GDP (%)[7/]	32.5	36.7	36.4	36.7	36.6	35.1	35.6	n.a.	n.a.
External debt (billion, U.S. dollars)	132.2	155.2	151.7	155.2	157.9	154.2	159.3	157.6	n.a.
Short-term (%)	41.2	44.3	42.3	44.3	43.1	43.0	41.3	40.4	n.a.
Long-term (%)	58.8	55.7	57.7	55.7	56.9	57.0	58.7	59.6	n.a.
International reserves / Short-term external debt (times)	3.2	2.9	3.1	2.9	3.2	3.1	3.1	3.2	n.a.

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2018		2019			2020		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
> 9	0	0	0	1	1	1	2	2
8-9	0	0	0	2	3	3	4	2
7-8	0	0	1	5	7	6	7	5
6-7	1	2	3	11	13	11	11	8
5-6	8	9	8	18	17	15	15	12
4-5	27	20	17	21	19	17	16	15
3-4	36	26	23	18	16	16	15	15
2-3	22	23	22	13	12	13	12	14
1-2	6	13	15	7	7	9	8	11
0-1	1	5	8	3	3	5	5	8
(-1)-0	0	1	3	1	1	2	3	5
(-2)-(-1)	0	0	1	0	0	1	1	2
(-3)-(-2)	0	0	0	0	0	0	0	1
< (-3)	0	0	0	0	0	0	0	1

Table: Probability distribution of headline inflation forecast

Percent	2018		2019			2020		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
> 4.0	0	0	0	0	1	1	2	2
3.5-4.0	0	0	1	1	2	1	2	3
3.0-3.5	0	1	2	2	3	3	4	4
2.5-3.0	0	3	4	4	6	5	6	7
2.0-2.5	1	8	7	7	8	8	9	9
1.5-2.0	8	16	12	10	11	10	11	11
1.0-1.5	23	20	15	13	12	12	12	11
0.5-1.0	30	19	15	14	12	12	12	11
0.0-0.5	22	15	14	13	11	11	11	10
(-0.5)-0.0	11	10	11	11	10	10	9	9
(-1.0)-(0.5)	3	5	8	9	8	8	8	7
(-1.5)-(1.0)	1	2	5	7	6	6	6	6
(-2.0)-(-1.5)	0	1	3	5	4	5	4	4
< -2.0	0	0	3	6	6	8	7	7

Table: Probability distribution of core inflation forecast

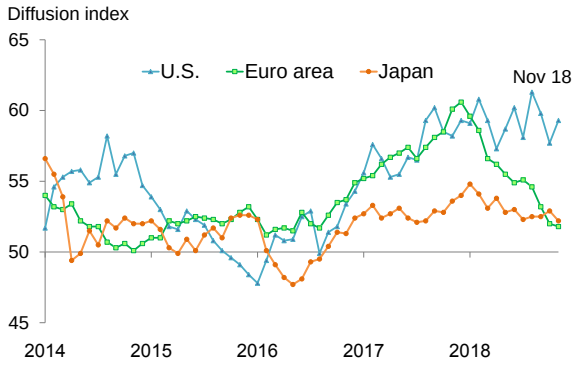
Percent	2018	2019				2020		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
> 4.0	0	0	0	0	0	0	0	0
3.5-4.0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	0	0	0
2.5-3.0	0	0	0	0	1	1	1	2
2.0-2.5	0	0	1	2	4	4	5	6
1.5-2.0	0	4	9	11	14	14	15	15
1.0-1.5	10	28	27	26	26	24	24	23
0.5-1.0	61	44	34	30	27	26	24	23
0.0-0.5	27	21	21	20	18	18	17	16
(-0.5)-0.0	1	4	7	8	8	9	9	9
(-1.0)-(0.5)	0	0	1	2	3	3	3	4
(-1.5)-(1.0)	0	0	0	0	1	1	1	1
(-2.0)-(-1.5)	0	0	0	0	0	0	0	0
< -2.0	0	0	0	0	0	0	0	0

4.2 Data pack

Global Economy

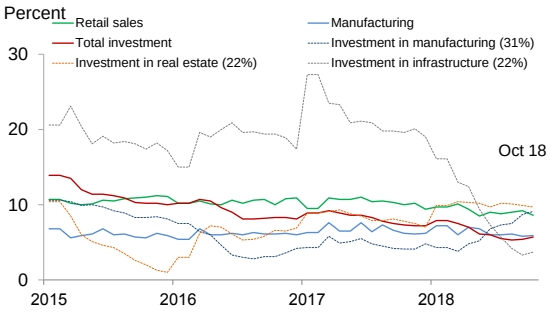
The U.S. and Japanese economies were expected to expand, while growth momentum stalled in the euro area due to domestic factors and the slowdown in global trade volume. The Chinese economy experienced a growth slowdown owing to the impact from policy to curb financial stability risks. Asian economies also slowed down due to tightening financial conditions, dampened confidence affected by the U.S. trade protectionism measures, as well as lower fiscal stimulus in some countries.

Manufacturing Purchasing Manager Index



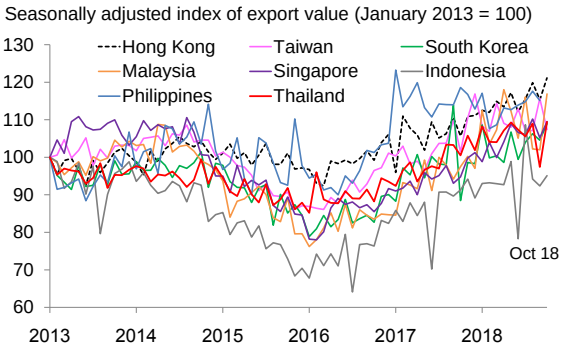
Sources: Bloomberg and Eurostat

China's economic indicators (Change from same period last year)



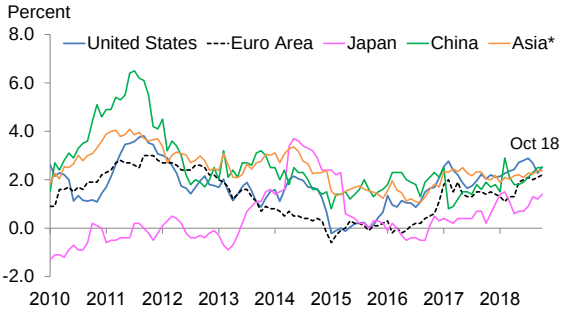
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners

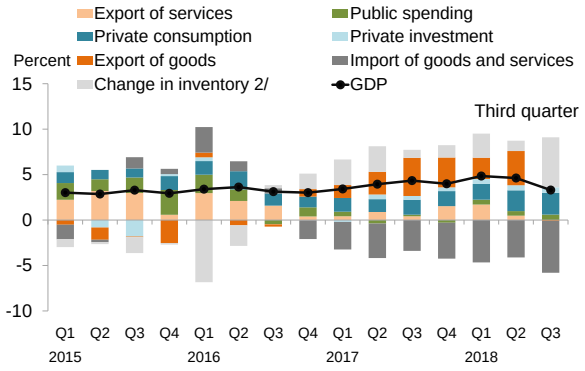


Note: * Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

Thai Economy

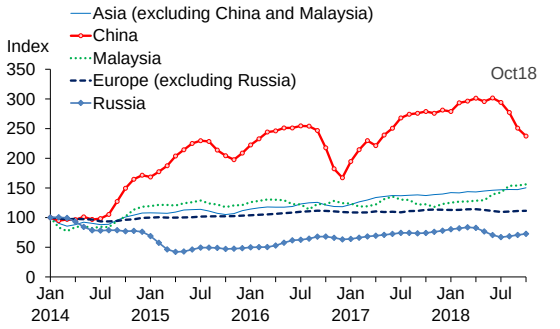
The Thai economy continued to gain further traction, mainly driven by private consumption whose growth accelerated in almost all product categories. On the other hand, merchandise exports growth slowed down, partly due to a temporary factor pertaining to storm situation in Thailand's trading partners and the impact from the U.S. trade protectionism measures. Exports of services exhibited slower growth, particularly for the number of Chinese tourists which declined following the Phuket tour boat sinking incident. Private investment growth accelerated especially in construction. Meanwhile, public expenditure continued to drive the economy.

Contribution to Thailand's GDP growth[1/]



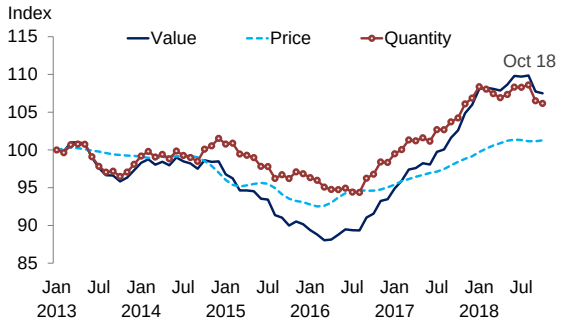
Note: [1/] Calculated by Chain Volume Measure method (CVM)
[2/] Change in inventory and statistical discrepancy
Source: Office of National Economic and Social Development Board, calculations by Bank of Thailand

Index of foreign tourists classified by nationality
(three-month moving average, seasonally adjusted; January 2014 = 100)



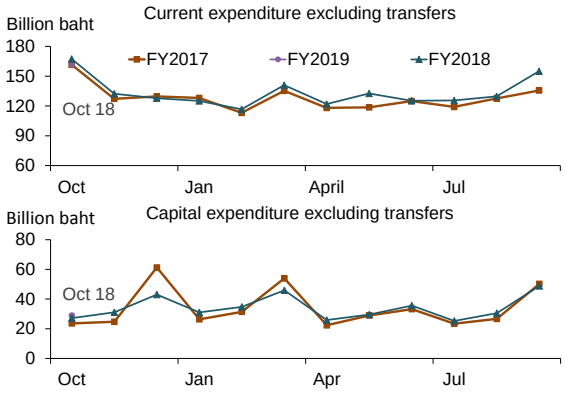
Source: Department of Tourism

Thai exports (excluding gold): value, price and quantity
(3-month moving average, seasonally adjusted; January 2013 = 100)



Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

Public spending by central government

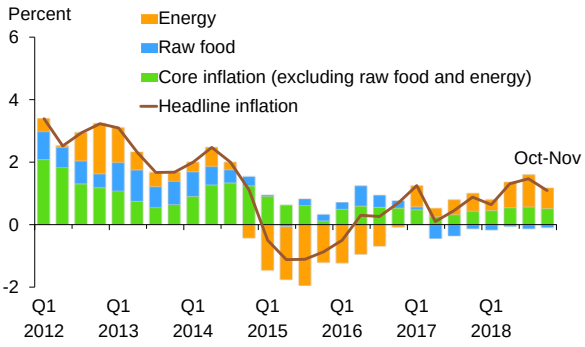


Source: Bureau of Budget, Fiscal Policy Office

Inflation

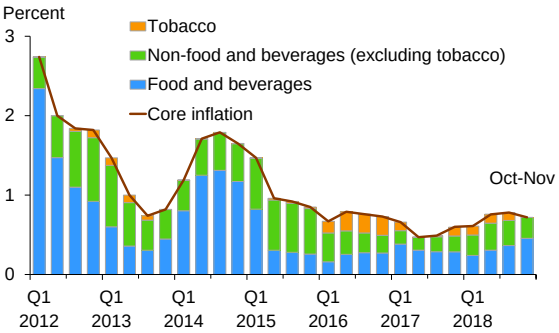
Headline inflation declined mainly on account of lower energy prices. Core inflation also decreased as the high base effect of excise tax increases on tobacco and alcoholic beverages was over. The short-term (one-year-ahead) inflation expectations according to the survey of businesses remained largely unchanged from the previous quarter.

Contribution to headline inflation



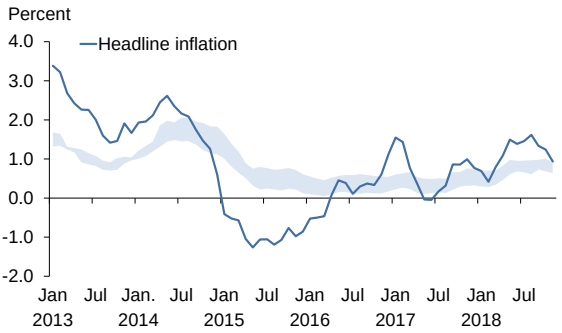
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

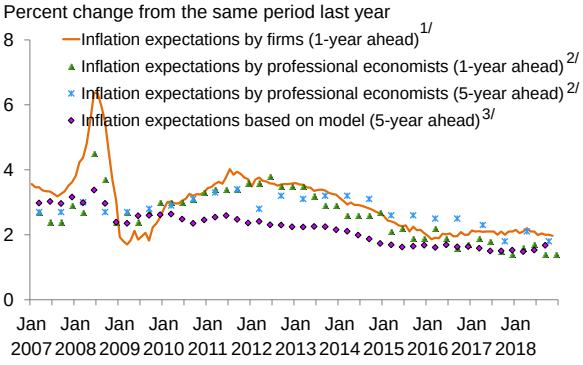
Underlying inflation indicators



Note: The field shows the highest and lowest outcomes among different measures of underlying inflation. The measures included are 1. **Asymmetric trim** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute price movements across categories of goods and services) and 3. **Core inflation excluding rent and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

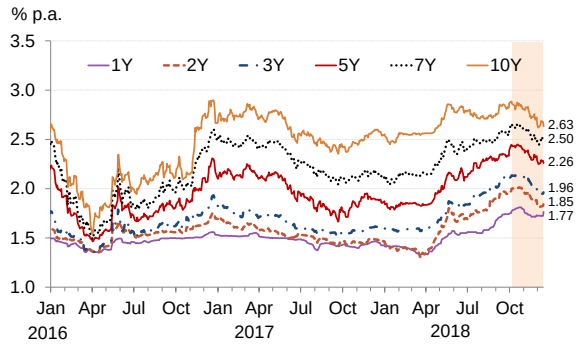


Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

Financial conditions

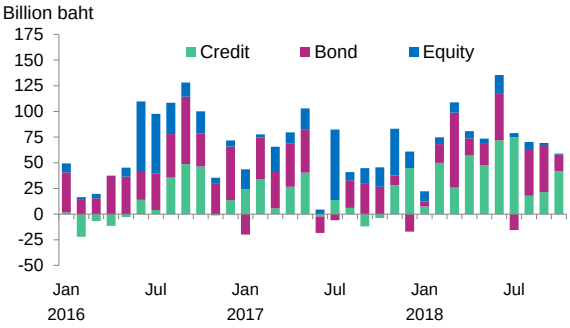
Short-term government bond yields rose in the beginning of the fourth quarter of 2018, partly on account of increased bond supply, before gradually declining owing to higher demand for Bank of Thailand bonds that were due to mature in December. Short-term yields later increased on account of market expectations of a policy rate hike prior to the 8th/2018 Monetary Policy Committee meeting. Medium-term and long-term government bond yields fell on account of domestic factors and market concerns over external fronts. Private credit to businesses and households continued to expand, consistent with economic conditions. The baht depreciated against the U.S. dollar due to market concerns over global economic uncertainty.

Government bond yields



Source: Thai Bond Market Association (Thai BMA) (data as of 18 December 2018)

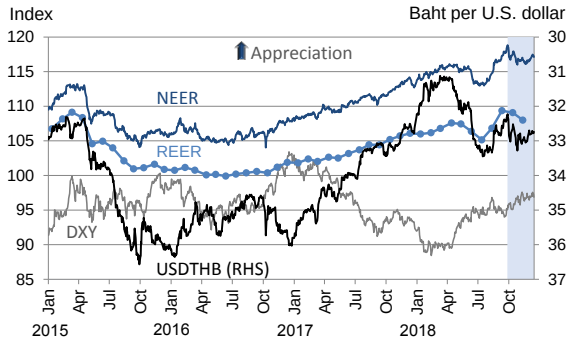
Total corporate financing by instrument*



Note: * Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding commercial banks, and newly issued equities.

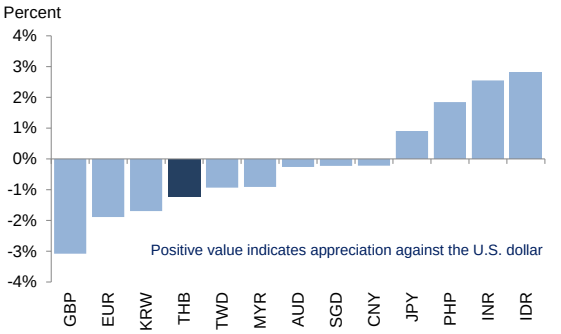
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and the Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of 18 September 2018)

Currency movements vis-a-vis the U.S. dollar (18 Dec 2018 compared to 30 Sep 2018)

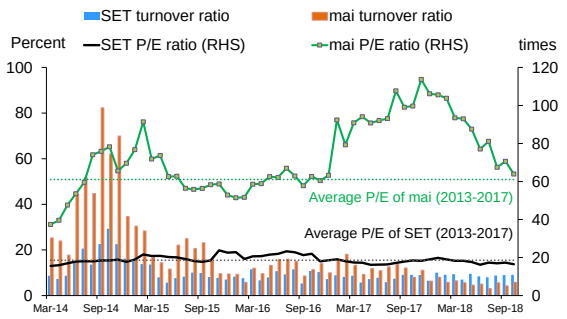


Source: Bank of Thailand and Reuters (data as of 18 December 2018)

Stability: financial markets

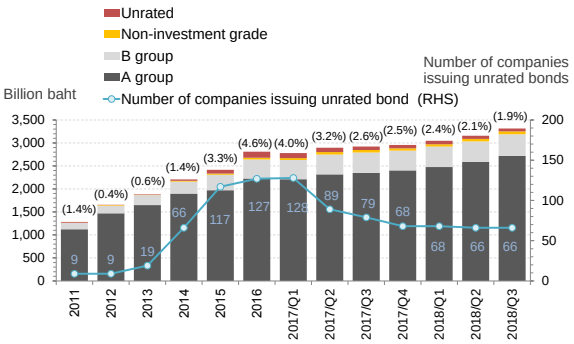
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand (SET) stayed slightly below the historical average, while that of the Market for Alternative Investment (mai) continued to trend down albeit remaining above the historical average. The share of unrated bonds issuance fell to 1.9 percent of total corporate bond outstanding in the third quarter of 2018.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of Sep 2018)

Corporate bonds outstanding

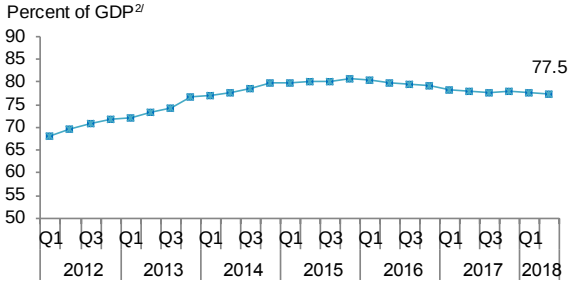


Note: ( ) represents percent of unrated bonds in total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: household sector

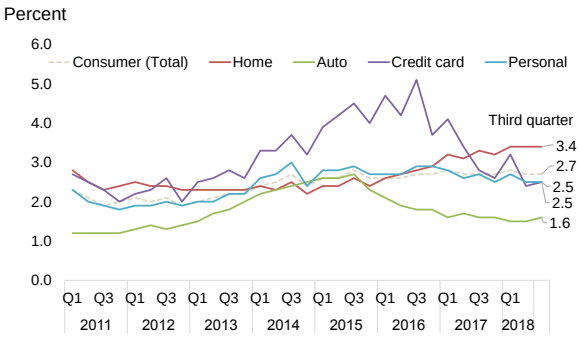
The ratio of household debt to GDP remained high despite a gradual decline. Credit quality of consumer loans overall remained stable at the end of the third quarter of 2018. Nonetheless, the NPL ratio for mortgage loans stayed at a high level while the NPL ratios increased for auto hire purchase and credit card loans.

Household debt[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
[3/] Household debt and GDP data are revised. This results in the different debt to GDP ratios compared to the last MPR.
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

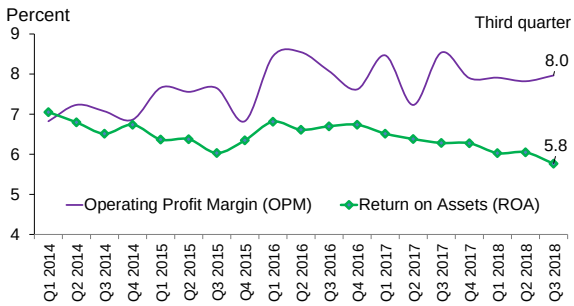


Source: Bank of Thailand

Stability: corporate sector

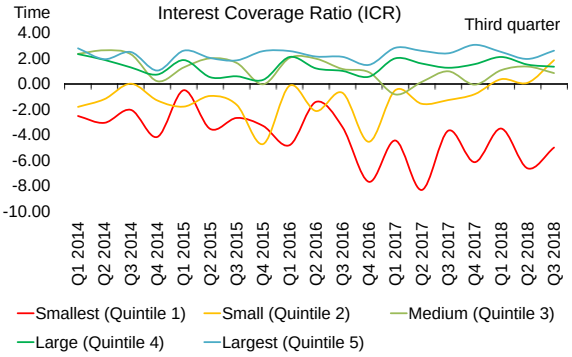
In the third quarter of 2018, stability of the corporate sector remained sound in line with continued economic expansion, as reflected in the operating profit margin and return on assets. Meanwhile, debt serviceability of small enterprises and businesses in some sectors remained vulnerable. The ratio of NPL among SMEs continued to rise, especially for those adversely affected by structural changes such as those in rice mills and retail trade sectors.

Operating Profit Margin (OPM) and Return on Assets (ROA)*



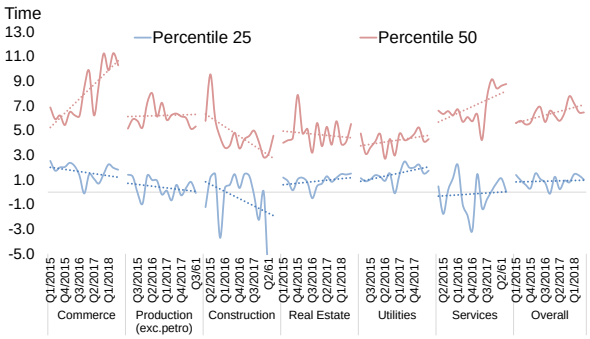
Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Debt serviceability at 25th percentile of each group of firm size



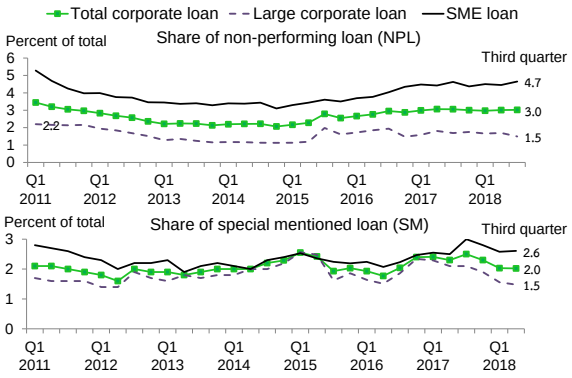
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Interest Coverage Ratio, classified by sectors



Note: * production exclude Petroleum and chemicals
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

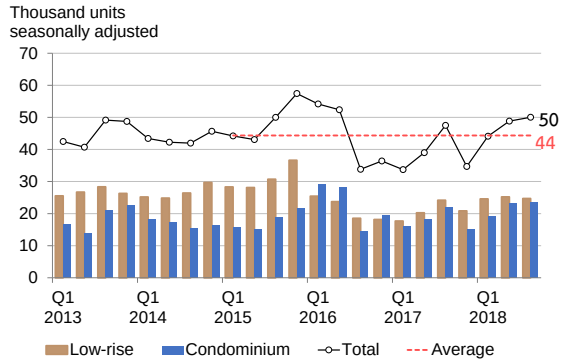


Source: Bank of Thailand

Stability: real estate

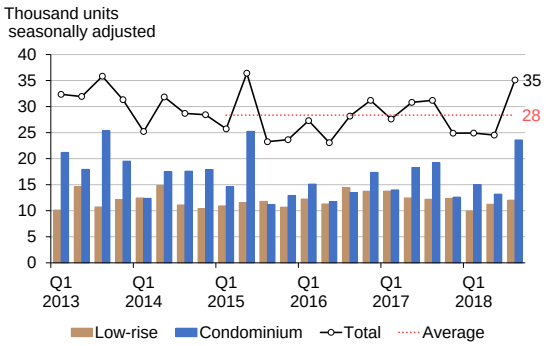
Demand for residences slightly increased, as reflected in the number of residential ownership transfers that continued to rise above the historical average. The supply also accelerated as seen by a large increase in newly opened residential units during the third quarter of 2018. This was particularly the case for condominiums as property developers focused on selling existing units rather than creating a new supply during the first half of the year. Property prices rose while land price declined, partly due to an expansion of property development projects in suburban areas.

Residential Transfer units in Bangkok and its vicinity



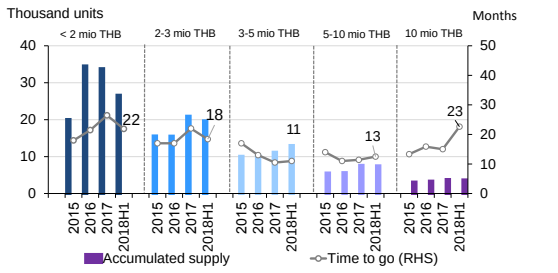
Note: *Average during 2014-2017
Source: Real Estate Information Center

New residential projects launched in Bangkok and its vicinity



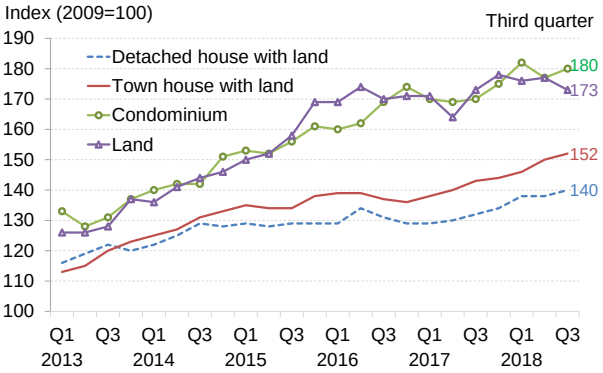
Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand
Note: *Average during 2014-2017

Condominium inventory in Bangkok and vicinity and 'Time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold out at the average sales rate per month (since projects launched) given no additional supply.
Source: AREA and calculation by Bank of Thailand

Real estate price indices

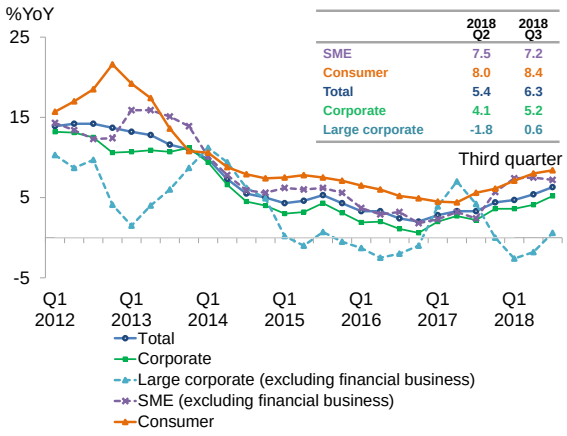


Source: Bank of Thailand

Stability: financial institutions

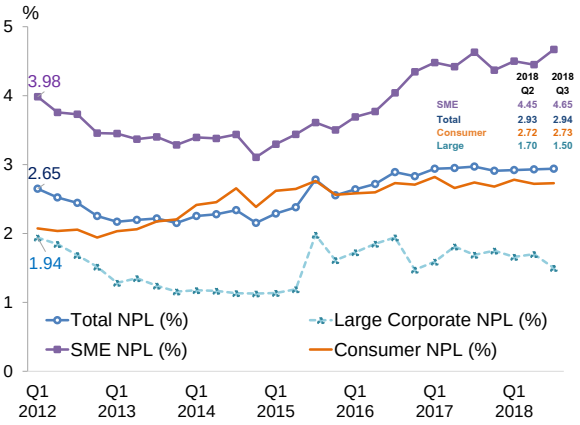
Financial institutions maintained strong financial positions, as reflected in high levels of capital buffers among commercial banks to cushion against risks should credit quality deteriorate. In the third quarter of 2018, commercial bank credits continued to grow for both business and consumer loans. The overall NPL ratio stabilized at a high level, especially for loans extended to SMEs that register a rapid growth.

Credit growth in the commercial bank system



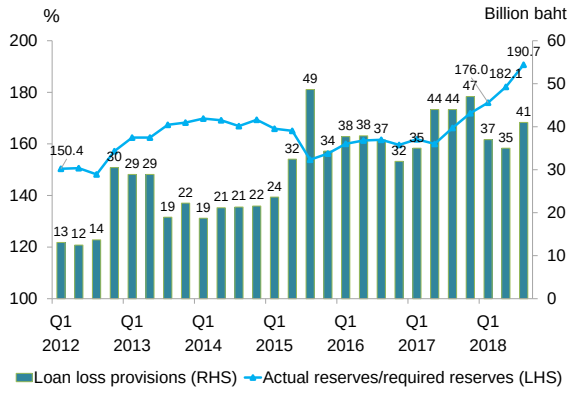
Source: Bank of Thailand

Non-performing loan (NPL)



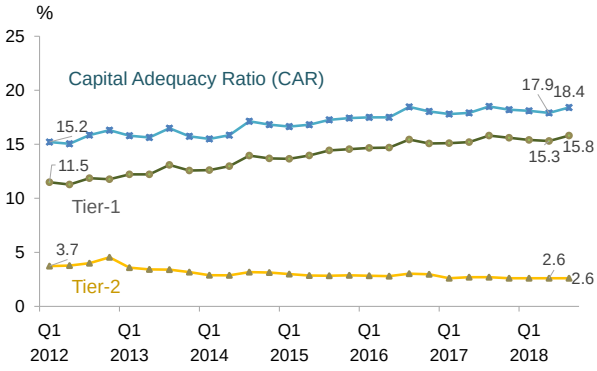
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

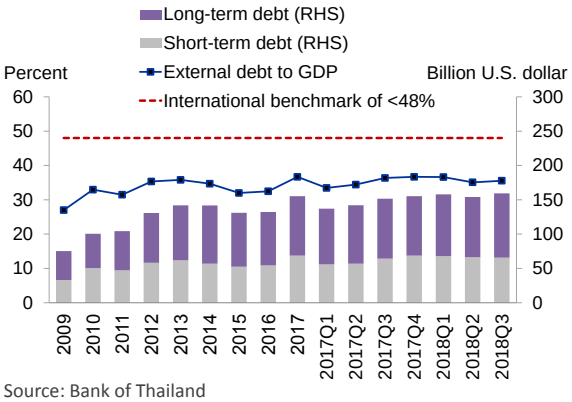


Source: Bank of Thailand

Stability: external position

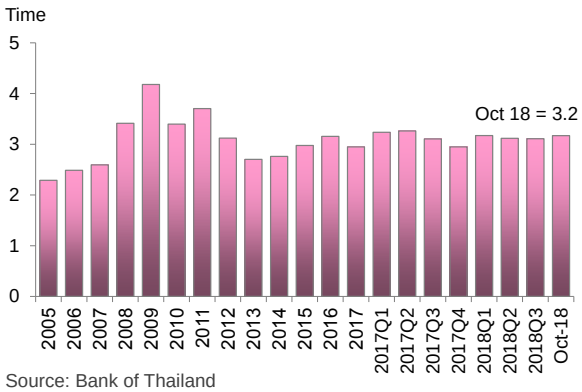
Thailand's external stability remained sound, as reflected in high levels of international reserves and a sustained current account surplus. Moreover, the ratio of external debt to GDP was below an international benchmark. This would help the Thai economy to be resilient against volatilities in global financial market.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

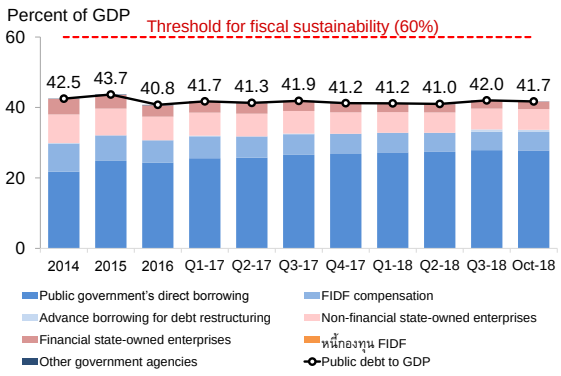


Source: Bank of Thailand

Stability: fiscal sector

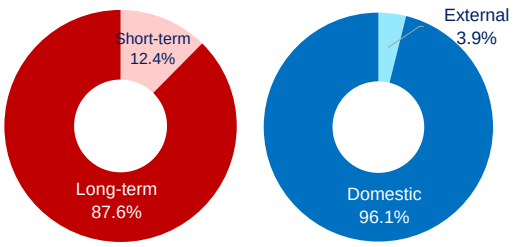
Fiscal stability remained sound. The ratio of public debt to GDP stayed below the sustainability threshold.

Public debt to GDP



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of October 2018



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office



Pursuing Sustainable
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